

การติดตั้ง Flutter สำหรับพัฒนา Mobile Application

Tools สำหรับการพัฒนา Application

Flutter

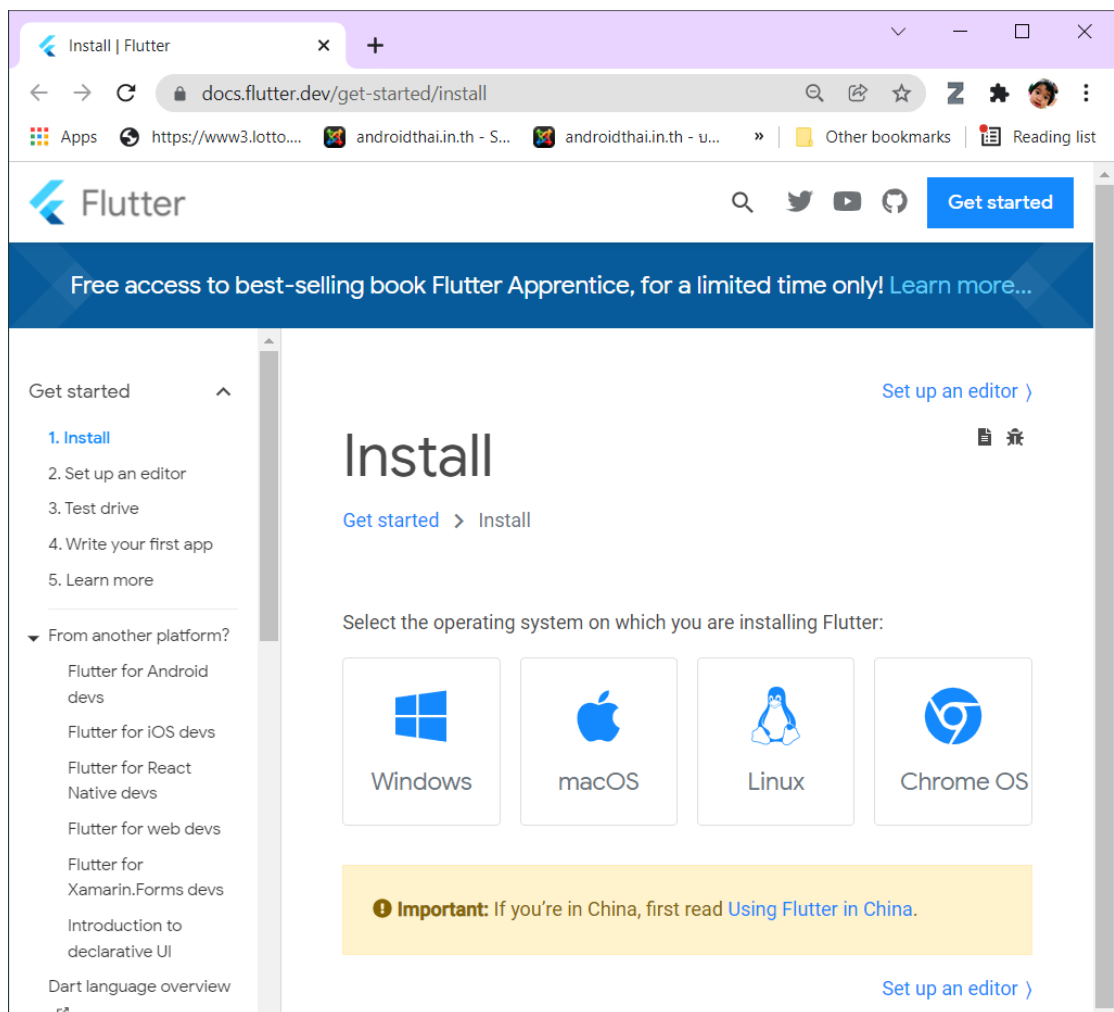
Android Studio

Android Emulator

VS Code และ Extension ที่เกี่ยวข้อง (flutter และภาษา dart)

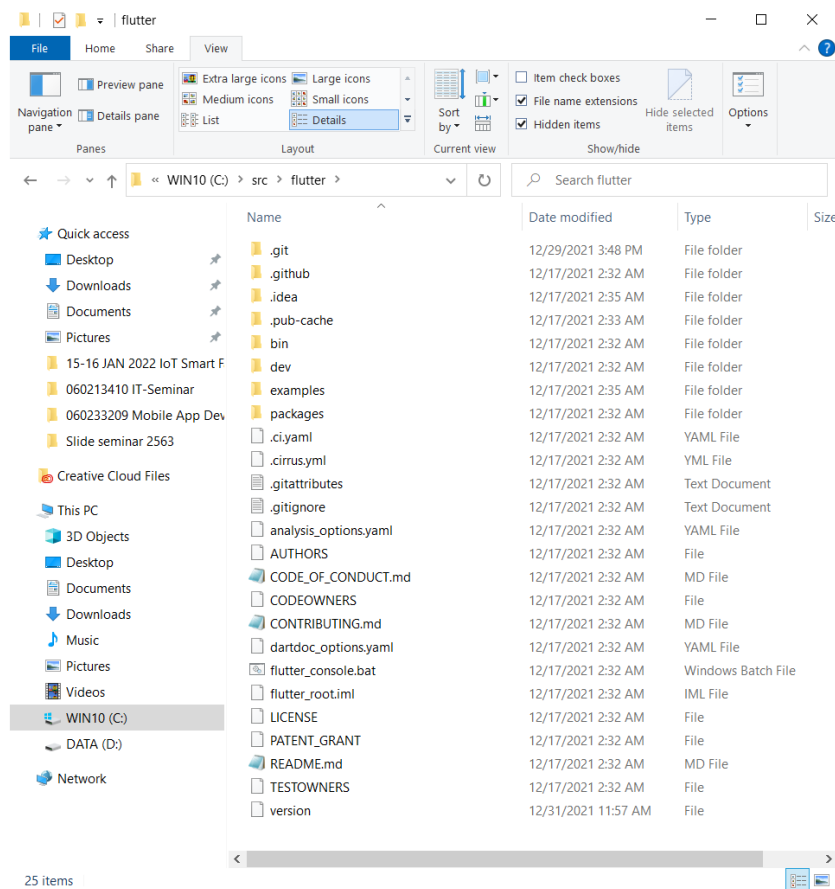
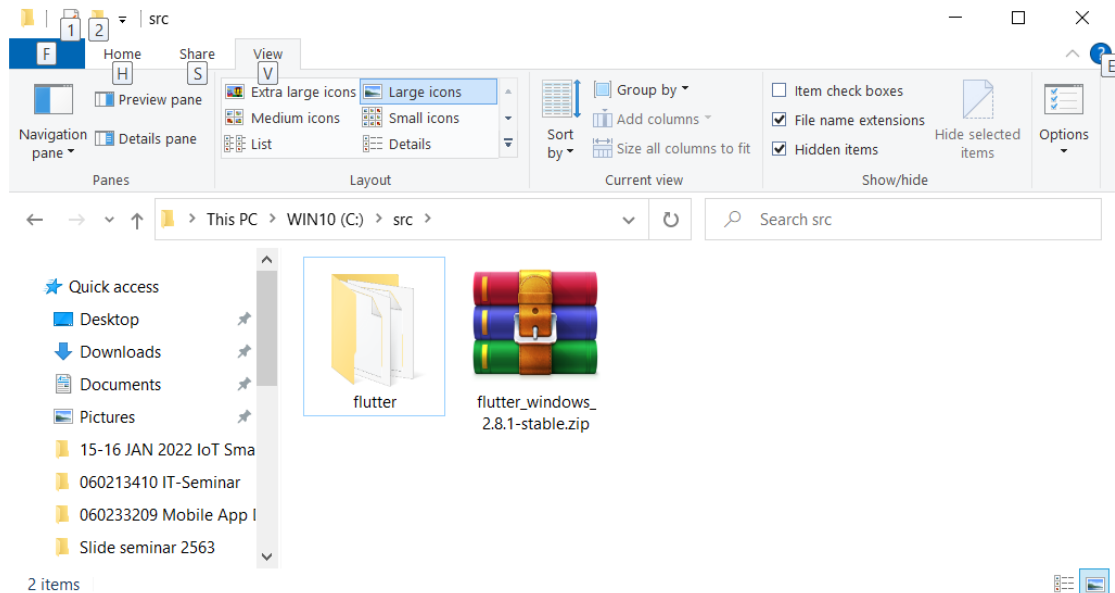
Flutter Installation

<https://docs.flutter.dev/get-started/install>

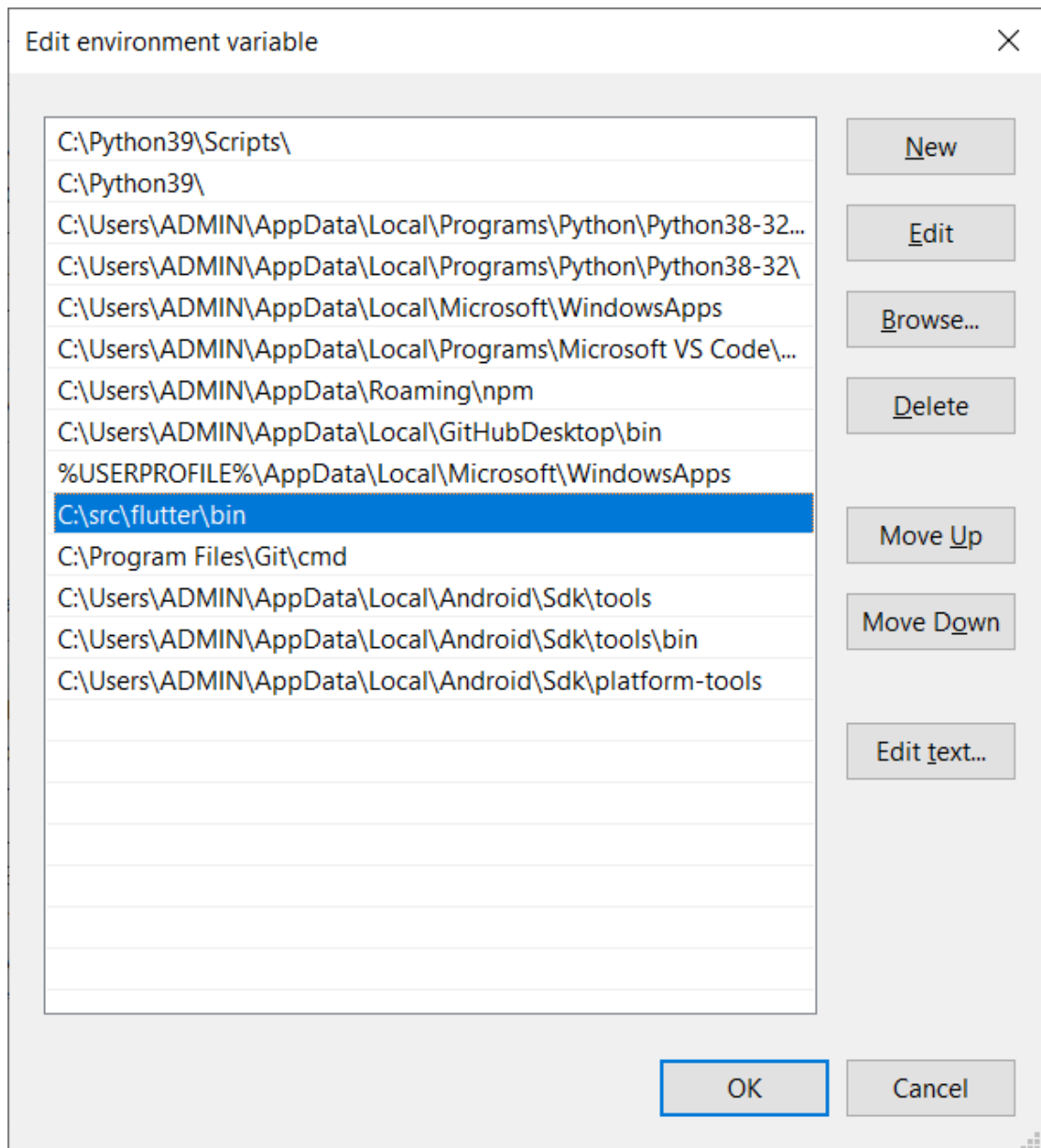


การติดตั้งใน Windows ทำตามขั้นตอนที่ <https://docs.flutter.dev/get-started/install/windows>

ได้ไฟล์ zip มาแล้วเปิดเข้าไปจะเจอกับโฟลเดอร์ flutter ให้แตกไฟล์ออกมาแล้วเอาไปไว้ในที่ที่ต้องการ เช่น C:\src\flutter



Set Path ที่ Edit environment variables



ตรวจสอบการติดตั้ง Flutter ด้วย command ต่อไปนี้

C:\Users\ADMIN>where flutter

```
Select Command Prompt
Microsoft Windows [Version 10.0.19043.1415]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ADMIN>where flutter
C:\src\flutter\bin\flutter
C:\src\flutter\bin\flutter.bat

C:\Users\ADMIN>
```

C:\Users\ADMIN>flutter เพื่อแสดง Common commands

```
Command Prompt

Common commands:

  flutter create <output directory>
    Create a new Flutter project in the specified directory.

  flutter run [options]
    Run your Flutter application on an attached device or in an emulator.

Usage: flutter <command> [arguments]

Global options:
-h, --help          Print this usage information.
-v, --verbose       Noisy logging, including all shell commands executed.
                    If used with "--help", shows hidden options. If used
                    with "flutter doctor", shows additional diagnostic
                    information. (Use "-vv" to force verbose logging in
                    those cases.)
-d, --device-id     Target device id or name (prefixes allowed).
--version           Reports the version of this tool.
--suppress-analytics Suppress analytics reporting when this command runs.

Available commands:

Flutter SDK
  bash-completion  Output command line shell completion setup scripts.
  channel          List or switch Flutter channels.
  config           Configure Flutter settings.
  doctor           Show information about the installed tooling.
  downgrade        Downgrade Flutter to the last active version for the current
                    channel.
  precache         Populate the Flutter tool's cache of binary artifacts.
  upgrade          Upgrade your copy of Flutter.

Project
  analyze          Analyze the project's Dart code.
  assemble         Assemble and build Flutter resources.
  build            Build an executable app or install bundle.
  clean            Delete the build/ and .dart_tool/ directories.
  create           Create a new Flutter project.
  drive            Run integration tests for the project on an attached device or
                    emulator.
  format           Format one or more Dart files.
  gen-l10n         Generate localizations for the current project.
  pub              Commands for managing Flutter packages.
  run              Run your Flutter app on an attached device.
  test             Run Flutter unit tests for the current project.

Tools & Devices
  attach           Attach to a running app.
  custom-devices  List, reset, add and delete custom devices.
  devices          List all connected devices.
  emulators        List, launch and create emulators.
  install          Install a Flutter app on an attached device.
  logs             Show log output for running Flutter apps.
  screenshot       Take a screenshot from a connected device.
  symbolize        Symbolize a stack trace from an AOT-compiled Flutter app.

Run "flutter help <command>" for more information about a command.
Run "flutter help -v" for verbose help output, including less commonly used options.

C:\Users\ADMIN>
```

ขั้นตอนการติดตั้ง Flutter เสร็จเรียบร้อยแล้ว

C:\Users\ADMIN>flutter doctor เพื่อตรวจสอบองค์ประกอบต่าง ๆ ของ flutter

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.18363.1139]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\KongRukSiam>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 1.22.3, on Microsoft Windows [Version 10.0.18363.1139], locale th-TH)

[!] Android toolchain - develop for Android devices (Android SDK version 30.0.1)
    ! Some Android licenses not accepted. To resolve this, run: flutter doctor --android-licenses
[!] Android Studio (not installed)
[!] Connected device
    ! No devices available

! Doctor found issues in 3 categories.
C:\Users\KongRukSiam>
```

*** ในกรณีที่ติดตั้งองค์ประกอบทุกอย่างครบสมบูรณ์แล้ว flutter doctor จะปรากฏดังภาพนี้

```
Command Prompt

C:\Users\ADMIN>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 2.8.1, on Microsoft Windows [Version 10.0.19043.1415], locale en-US)
[✓] Android toolchain - develop for Android devices (Android SDK version 32.0.0)
[✓] Chrome - develop for the web
[✓] Android Studio (version 2020.3)
[✓] VS Code (version 1.63.2)
[✓] Connected device (2 available)

• No issues found!
C:\Users\ADMIN>
```

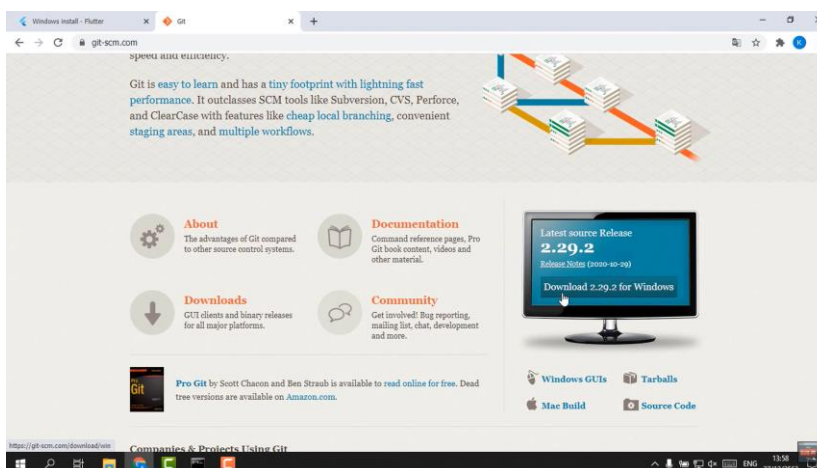
*กรณี flutter doctor ใช้งานไม่ได้

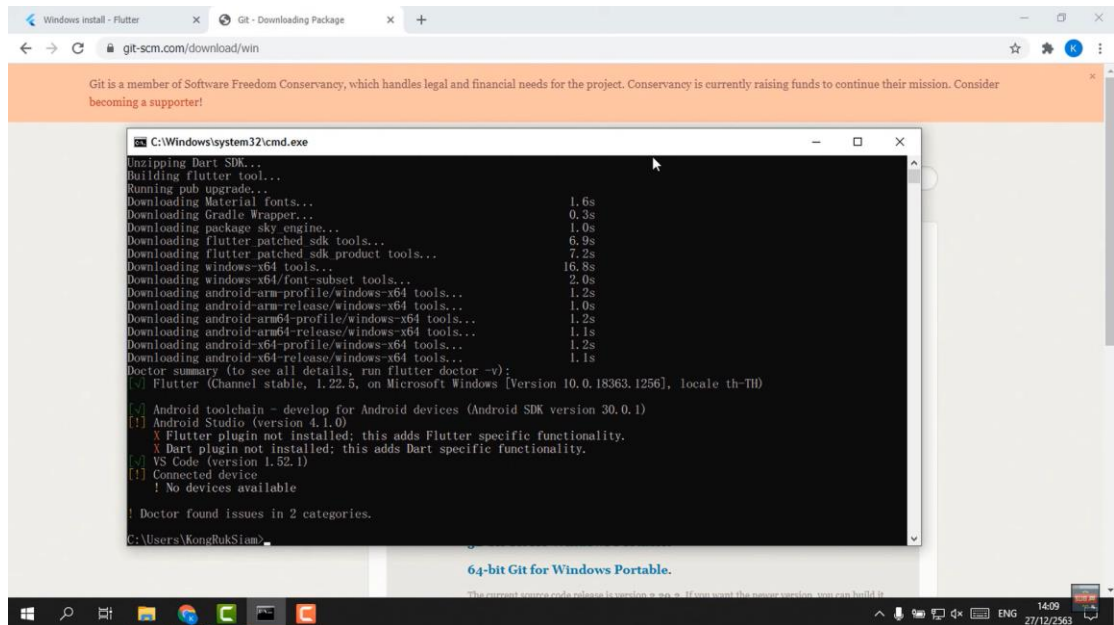
```
Select C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.18363.1256]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\KongRukSiam>flutter doctor
Error: Unable to find git in your PATH.

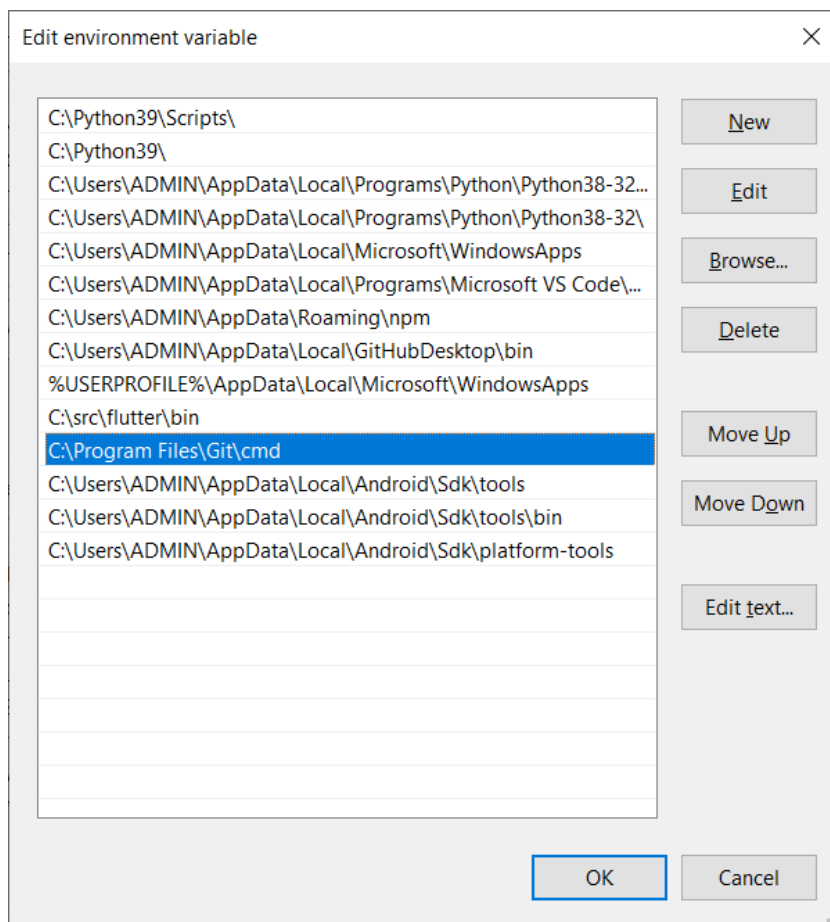
C:\Users\KongRukSiam>
```

แก้ไขโดยการติดตั้ง Git





หากยังไม่สามารถใช้งานได้ ให้ทำการ set path ของ Git

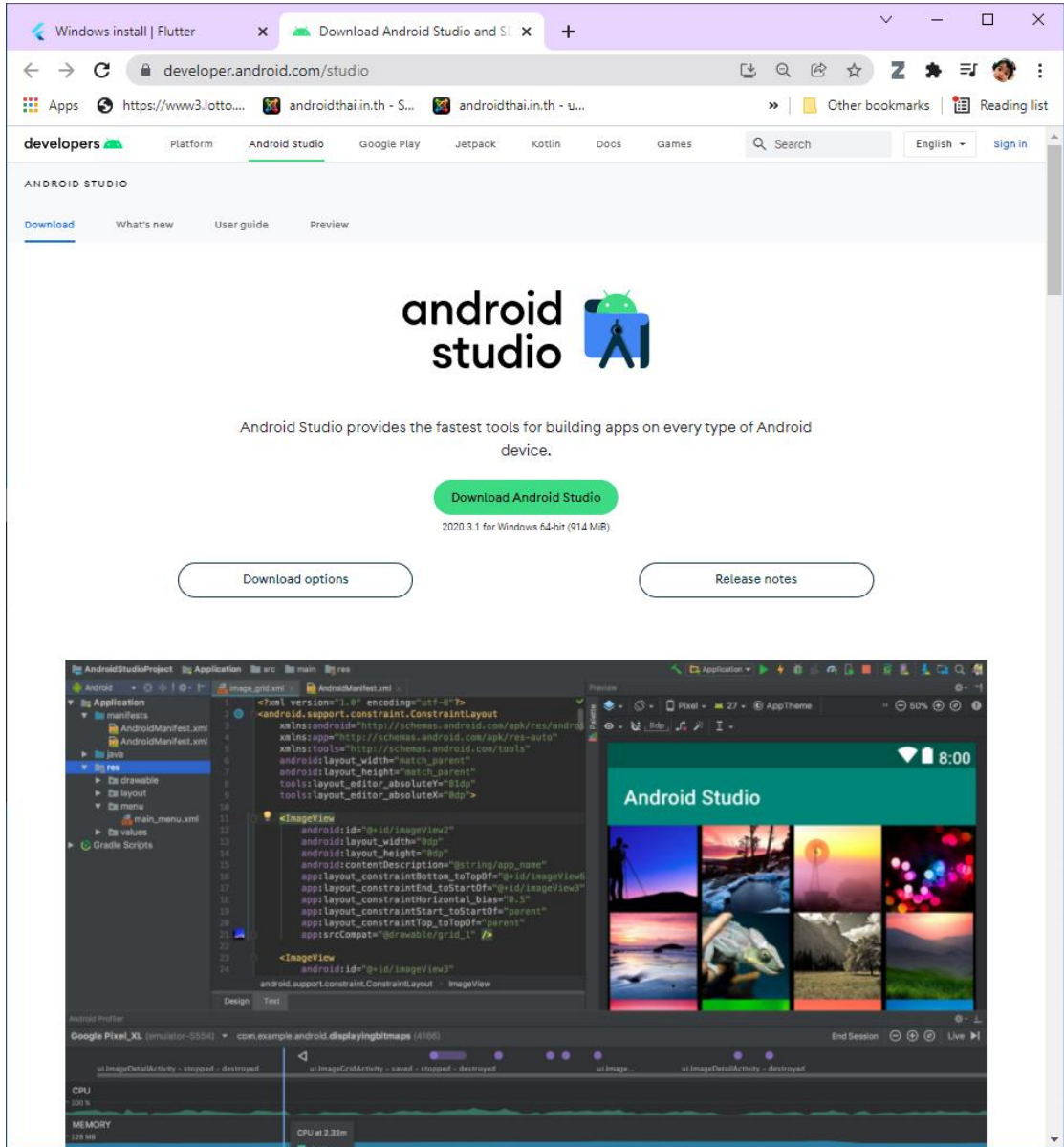


*ทดสอบคำสั่ง flutter doctor อีกครั้ง

Android Studio Installation

Install Android Studio

1. Download and install Android Studio.



Windows install | Flutter

Download Android Studio and SE

+

← → ↺

developer.android.com/studio

🔍 🔖 ⚙️ 👤 ⋮

📱 Apps

🌐 https://www3.lotto...

🌐 androidthai.in.th - S...

🌐 androidthai.in.th - u...

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📖 Reading list

developers

ANDROID STUDIO

Download

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13. Changes to the License Agreement

13.1 Google may make changes to the License Agreement as it distributes new versions of the SDK. When these changes are made, Google will make a new version of the License Agreement available on the website where the SDK is made available.

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☒ I have read and agree with the above terms and conditions

Download Android Studio 2020.3.1 for Windows

android-studio-2020.3.1.26-windows.exe

Android Studio Setup



Android Studio

Welcome to Android Studio Setup

Setup will guide you through the installation of Android Studio.

It is recommended that you close all other applications before starting Setup. This will make it possible to update relevant system files without having to reboot your computer.

Click Next to continue.

< Back

Next >

Cancel

Android Studio Setup



Choose Components

Choose which features of Android Studio you want to install.

Check the components you want to install and uncheck the components you don't want to install. Click Next to continue.

Select components to install:

☒ Android Studio

☒ Android Virtual Device

Space required: 2.5GB

Description

Position your mouse over a component to see its description.

< Back

Next >

Cancel

Android Studio Setup



Configuration Settings
Install Locations

Android Studio Installation Location

The location specified must have at least 500MB of free space.
Click Browse to customize:

< Back

Next >

Cancel

Android Studio Setup



Choose Start Menu Folder
Choose a Start Menu folder for the Android Studio shortcuts.

Select the Start Menu folder in which you would like to create the program's shortcuts. You can also enter a name to create a new folder.

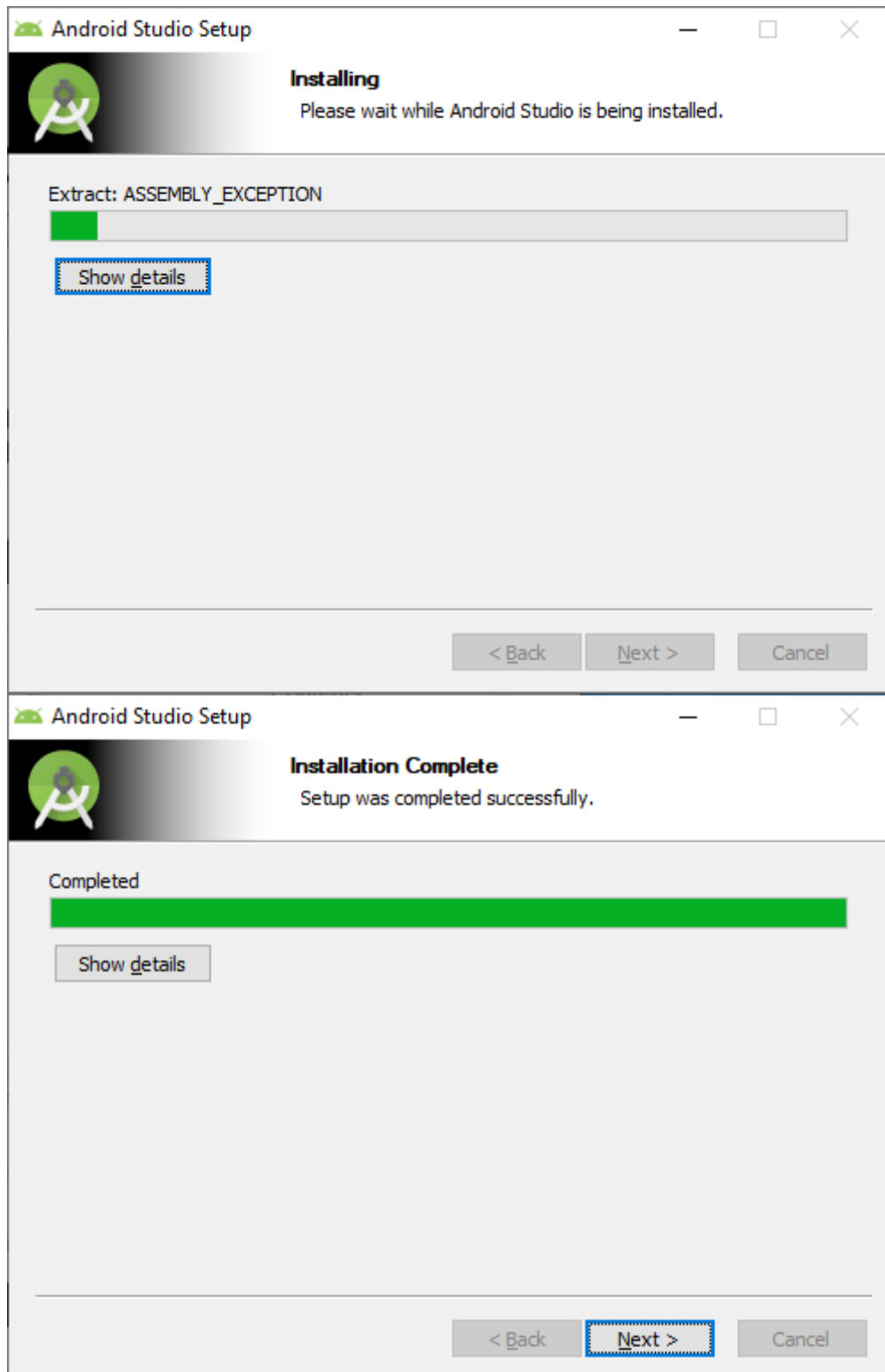
Accessibility
Accessories
Administrative Tools
AnyDesk
Autodesk
Cisco Packet Tracer
Git
Maintenance
SketchUp 2016
StartUp
System Tools

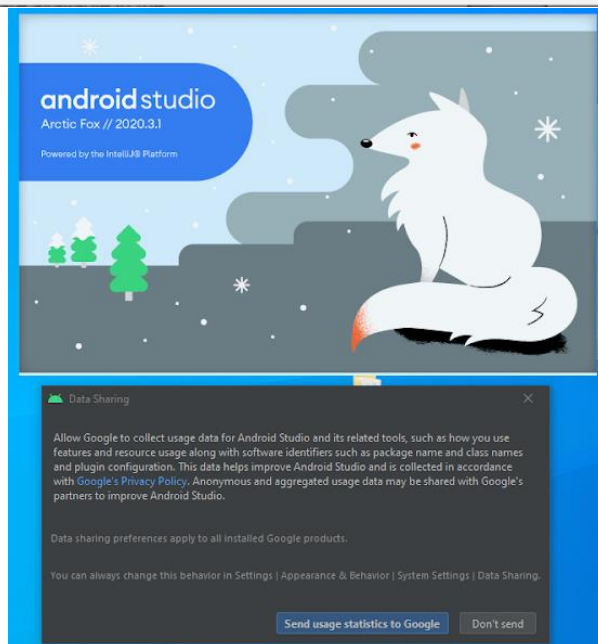
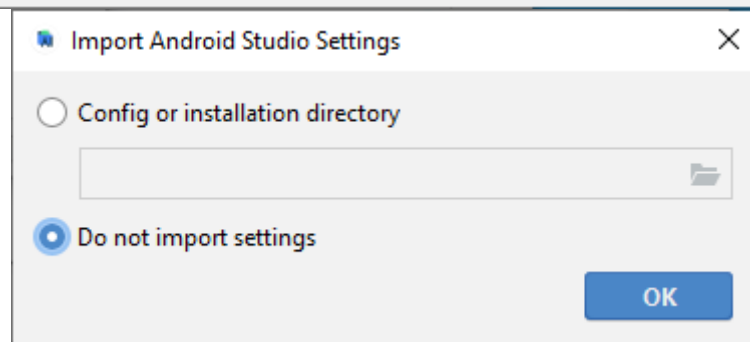
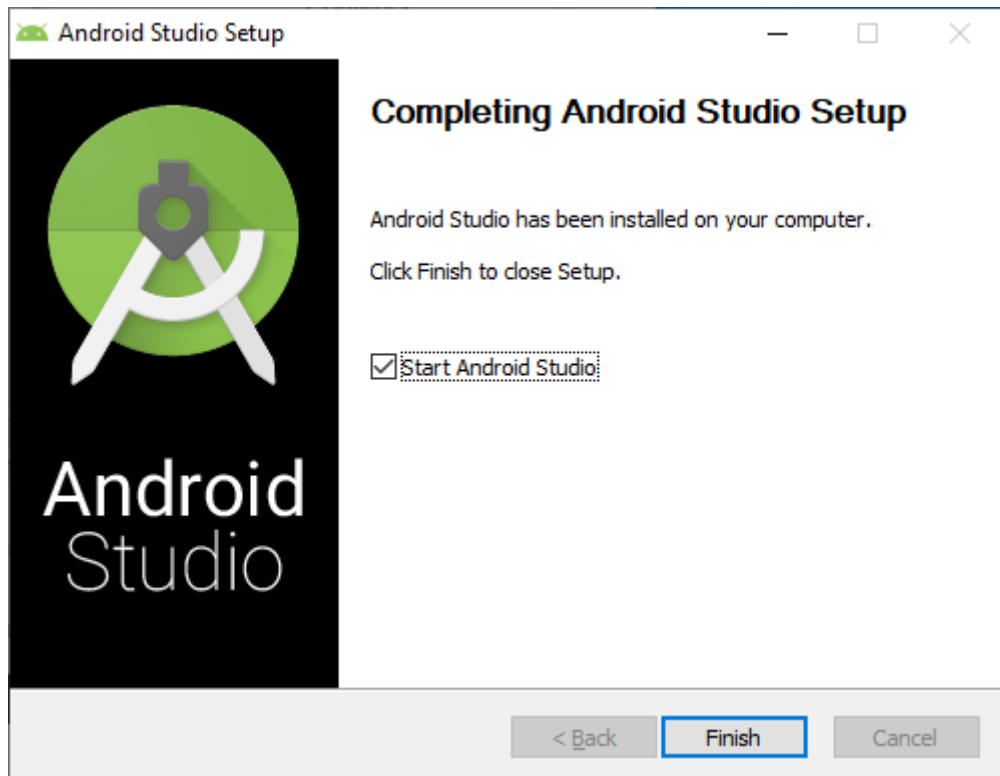
☐ Do not create shortcuts

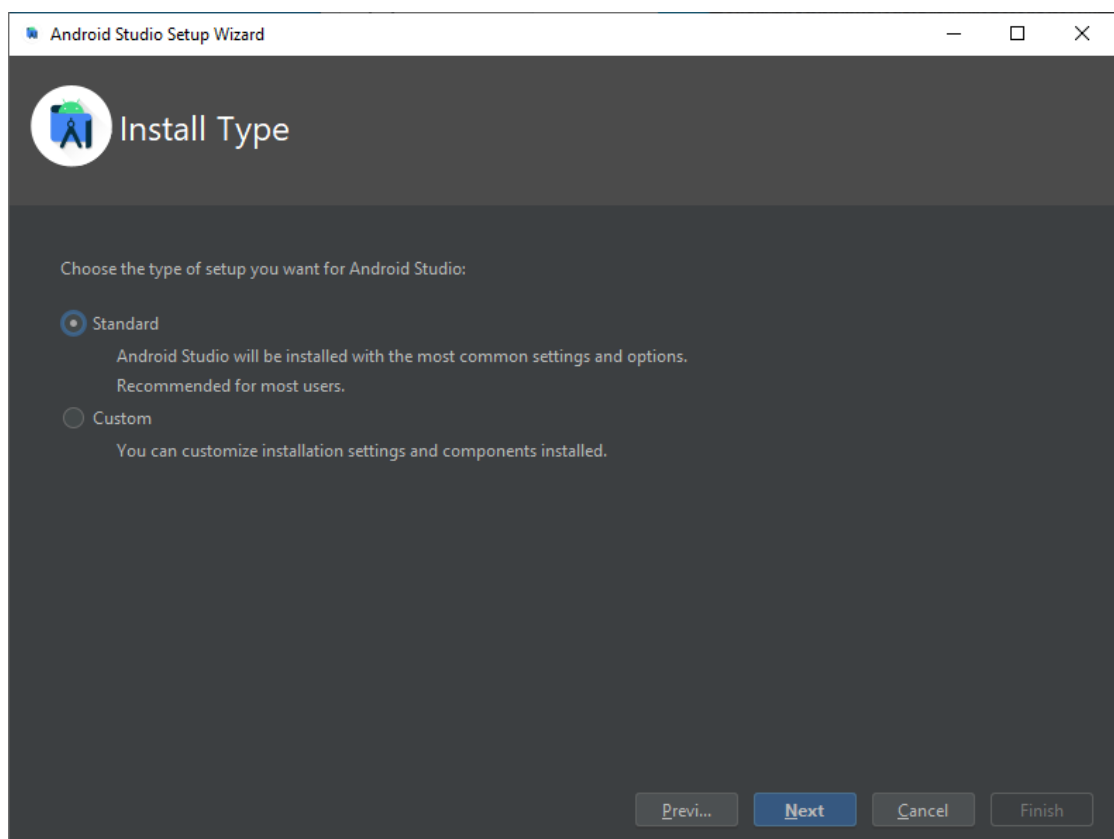
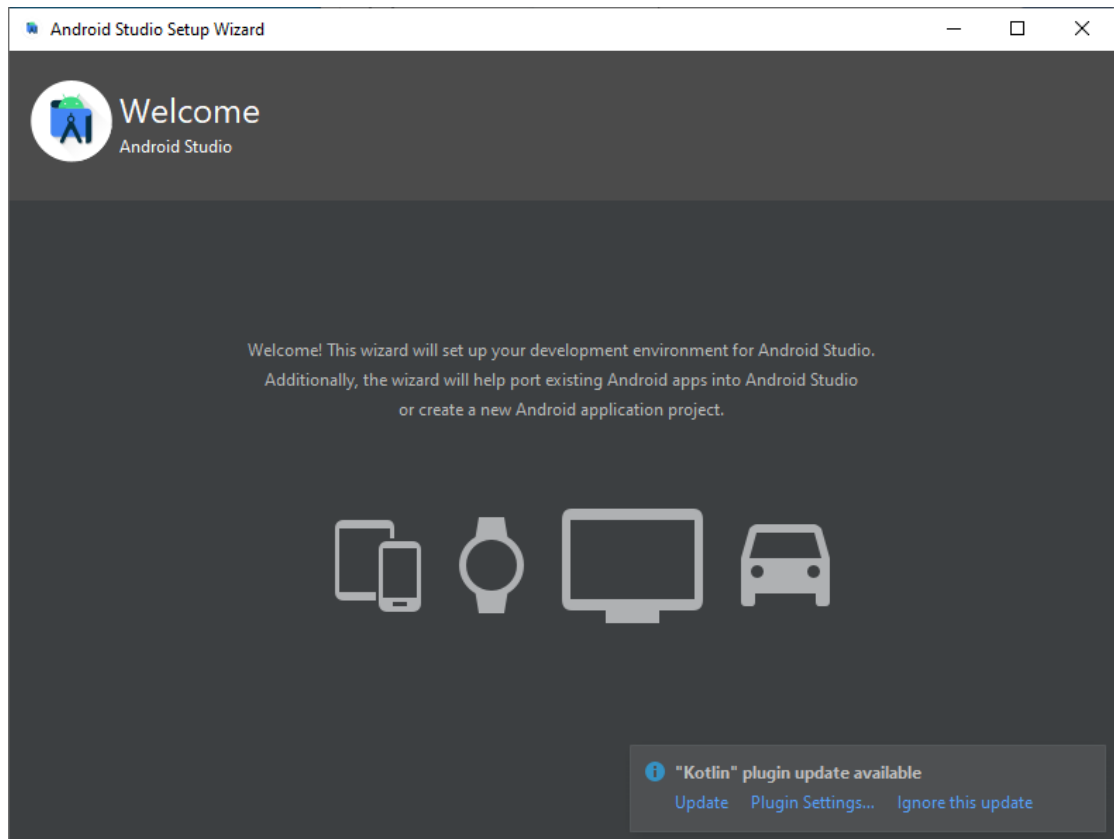
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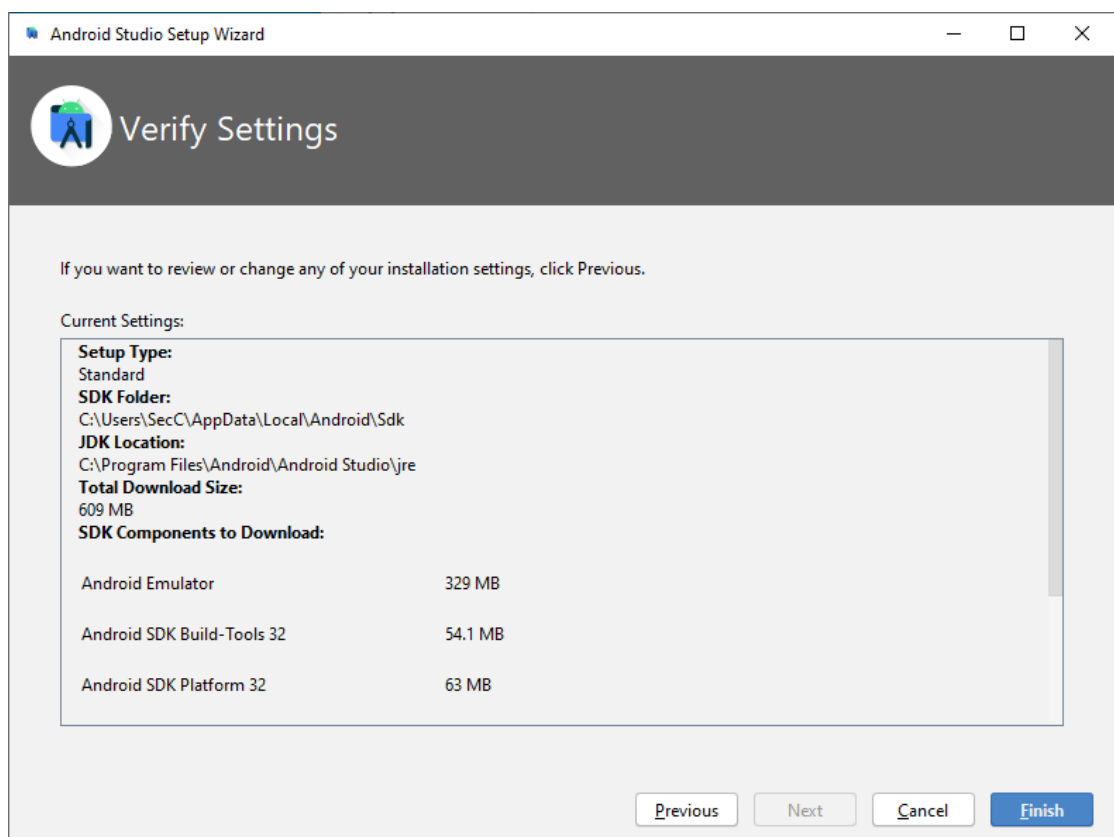
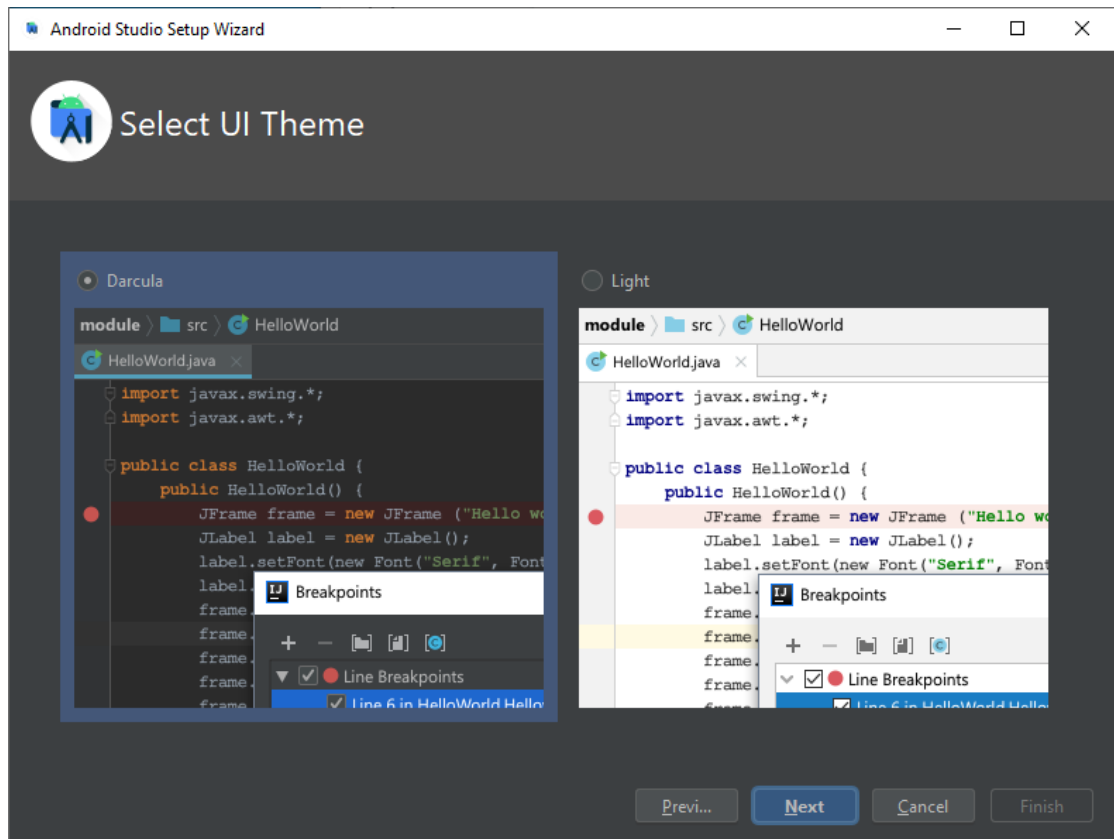
Install

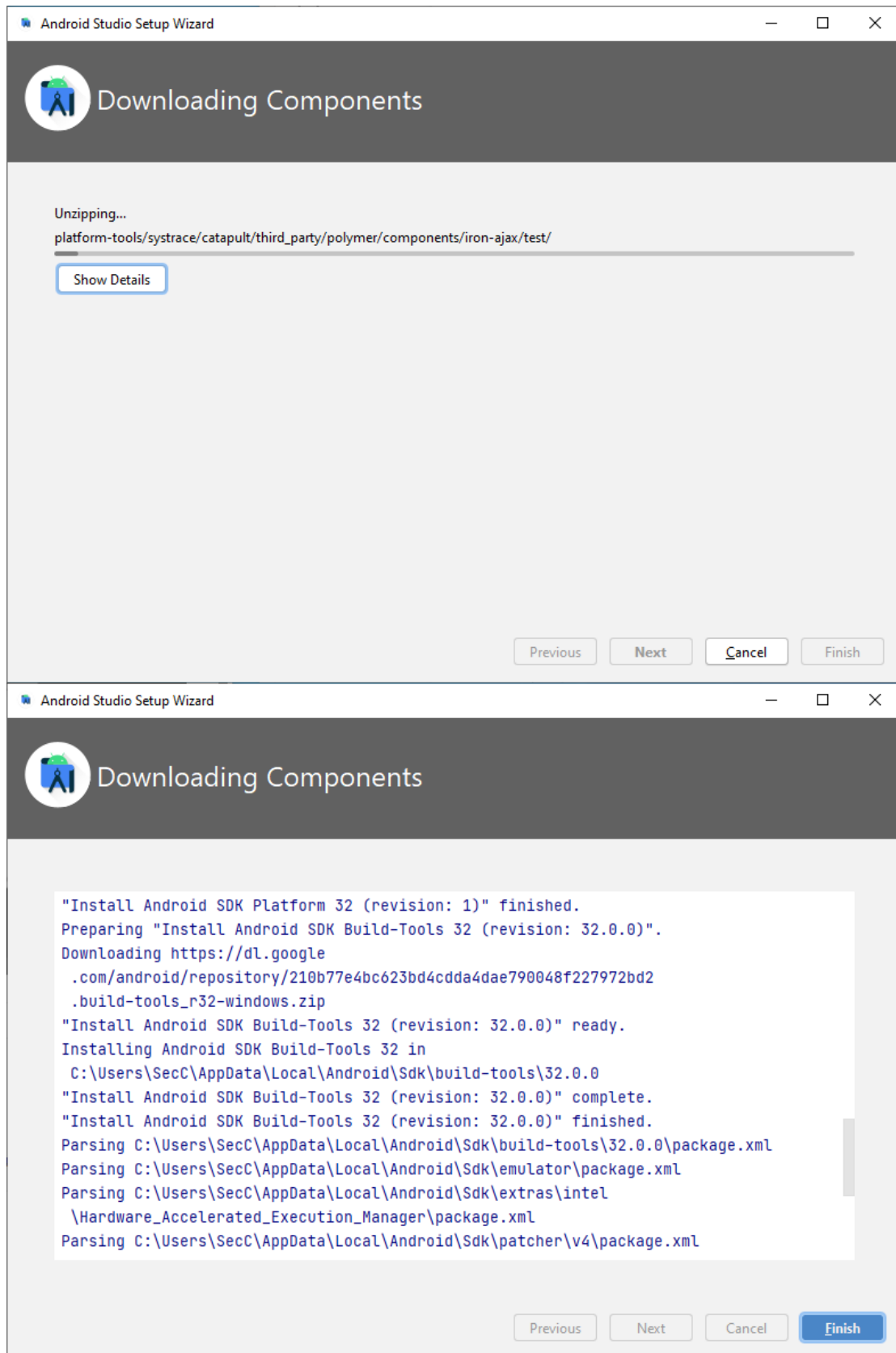
Cancel



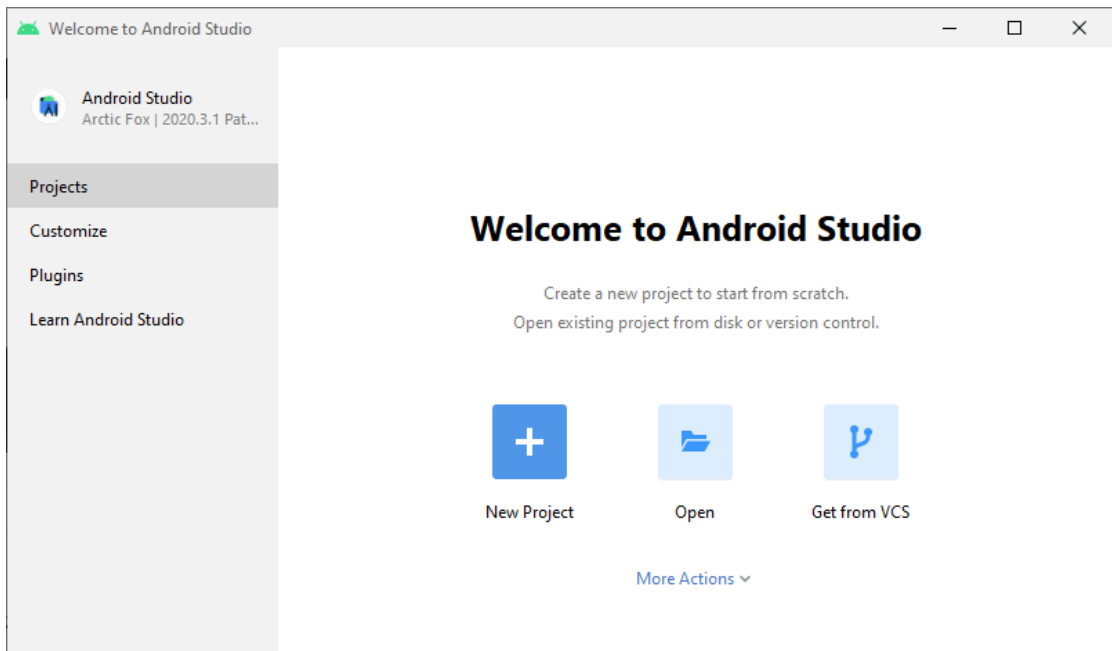




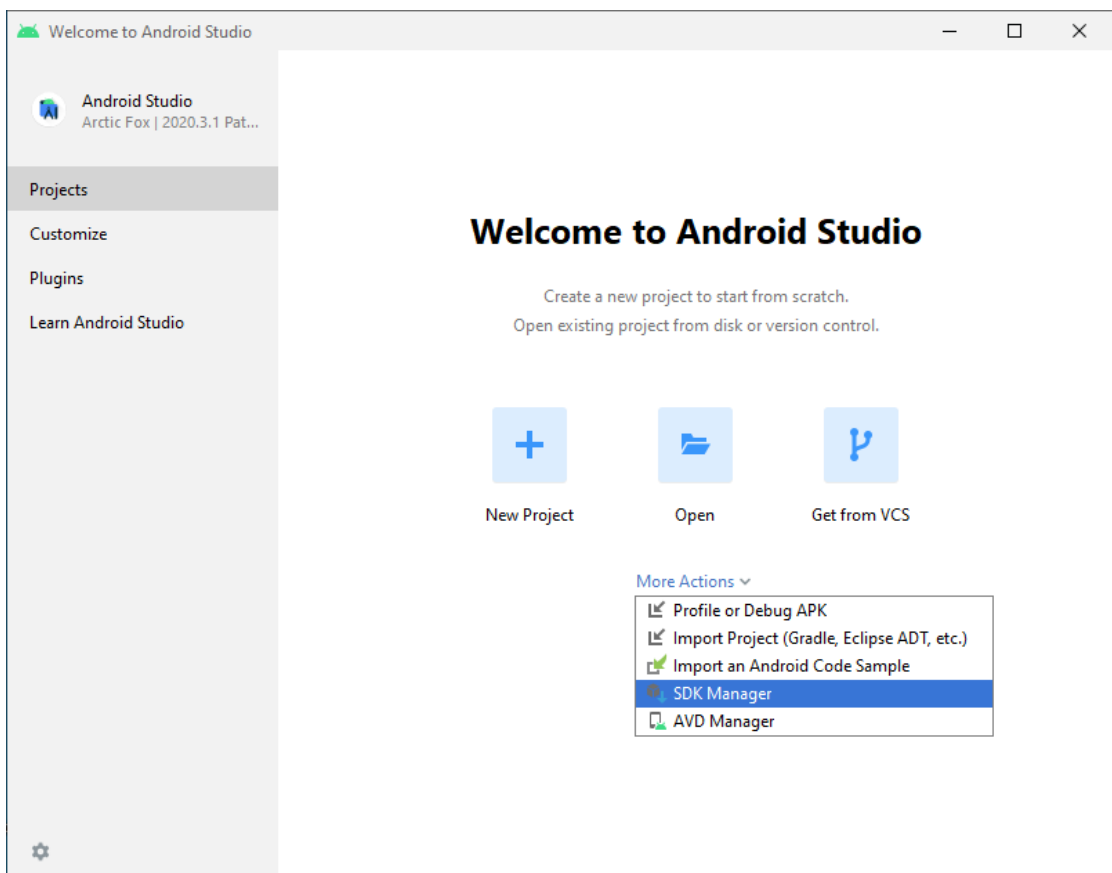


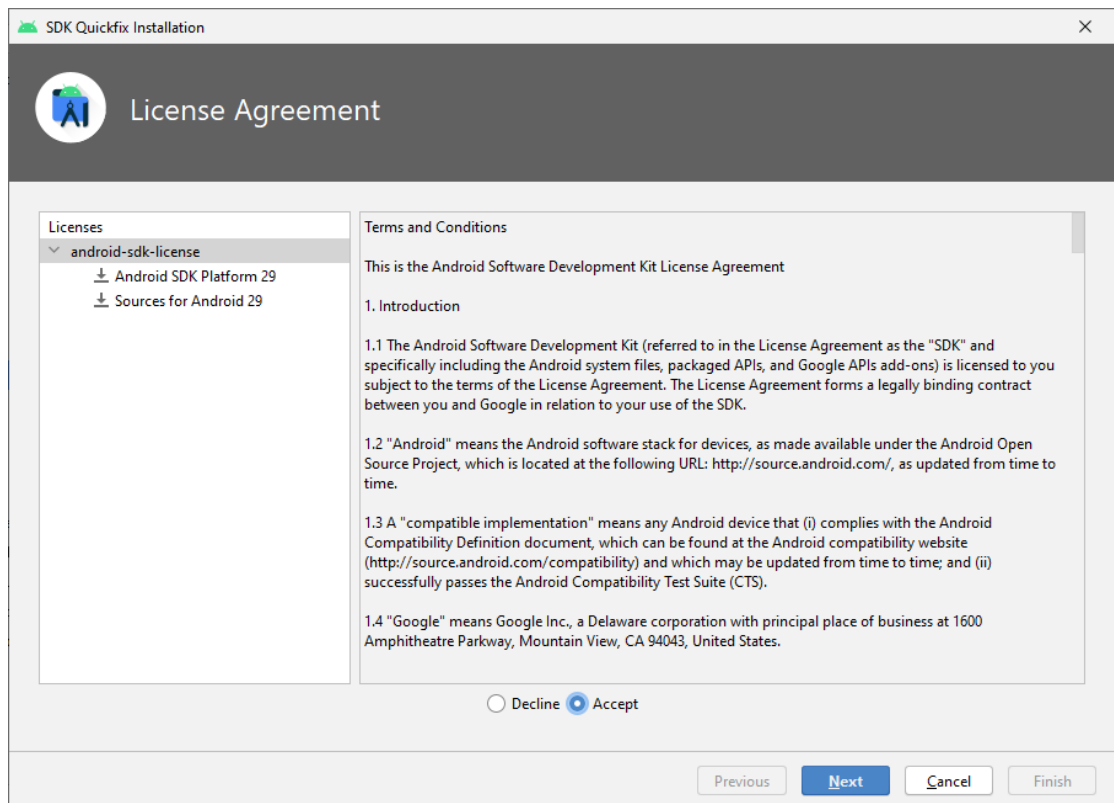
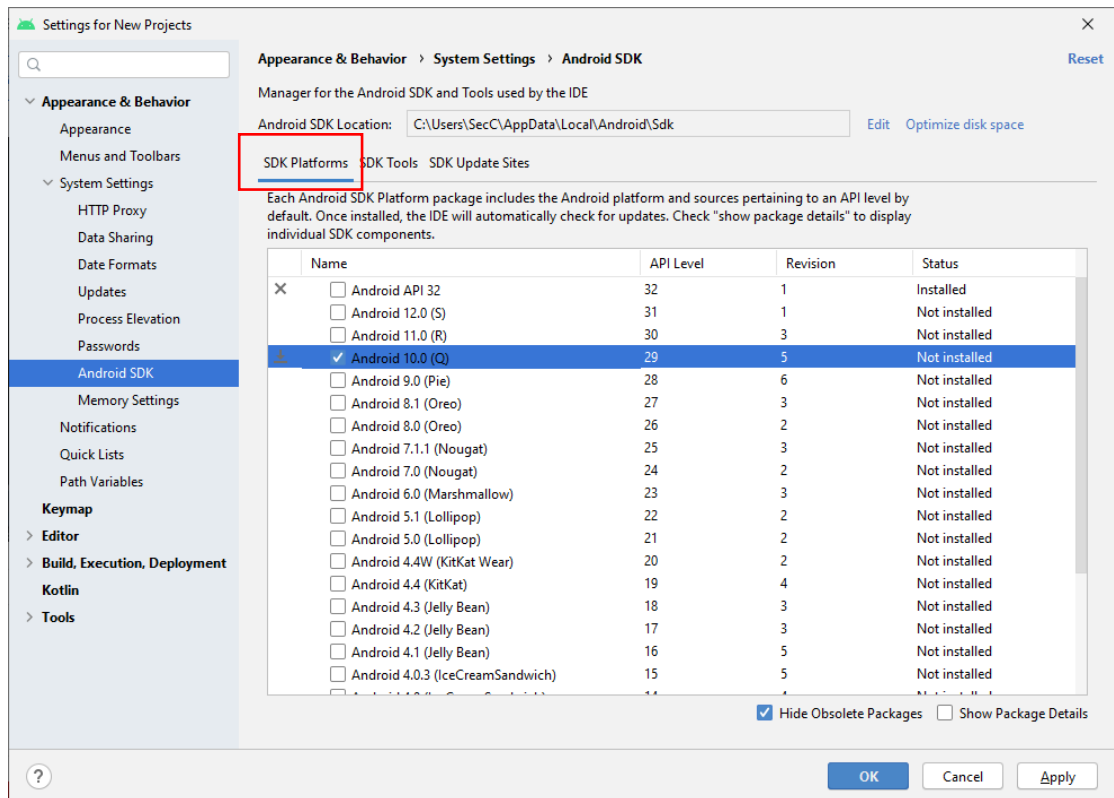


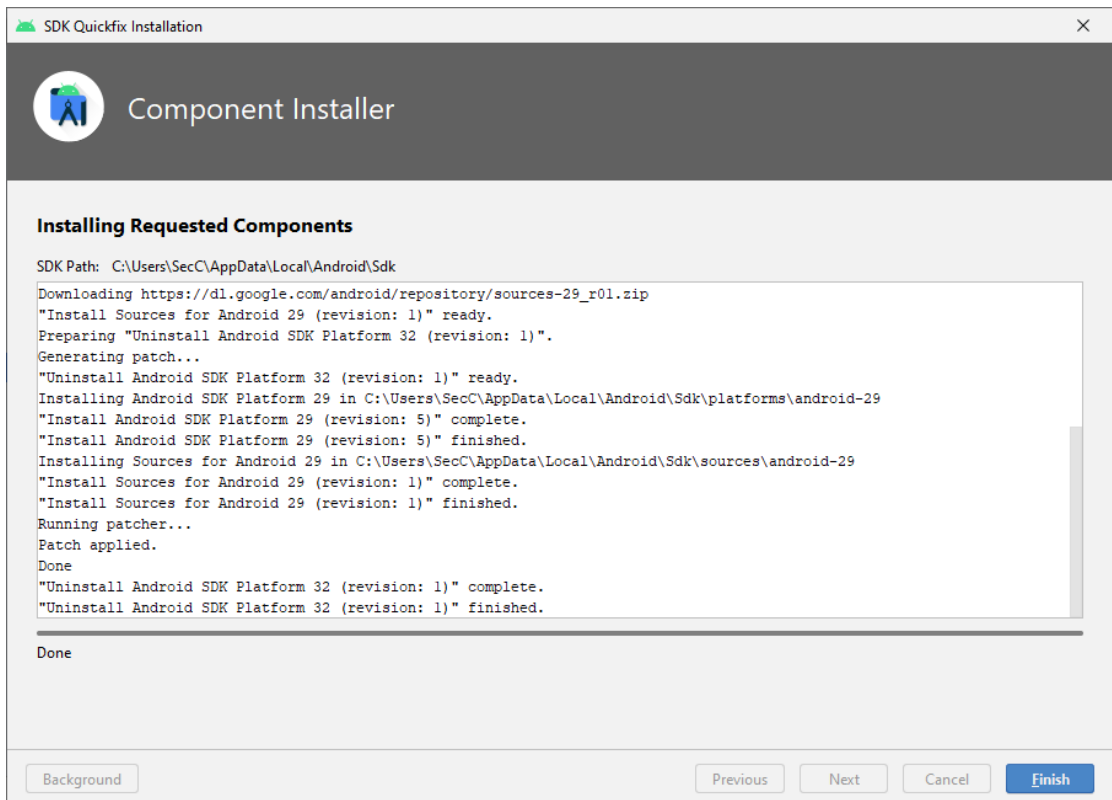
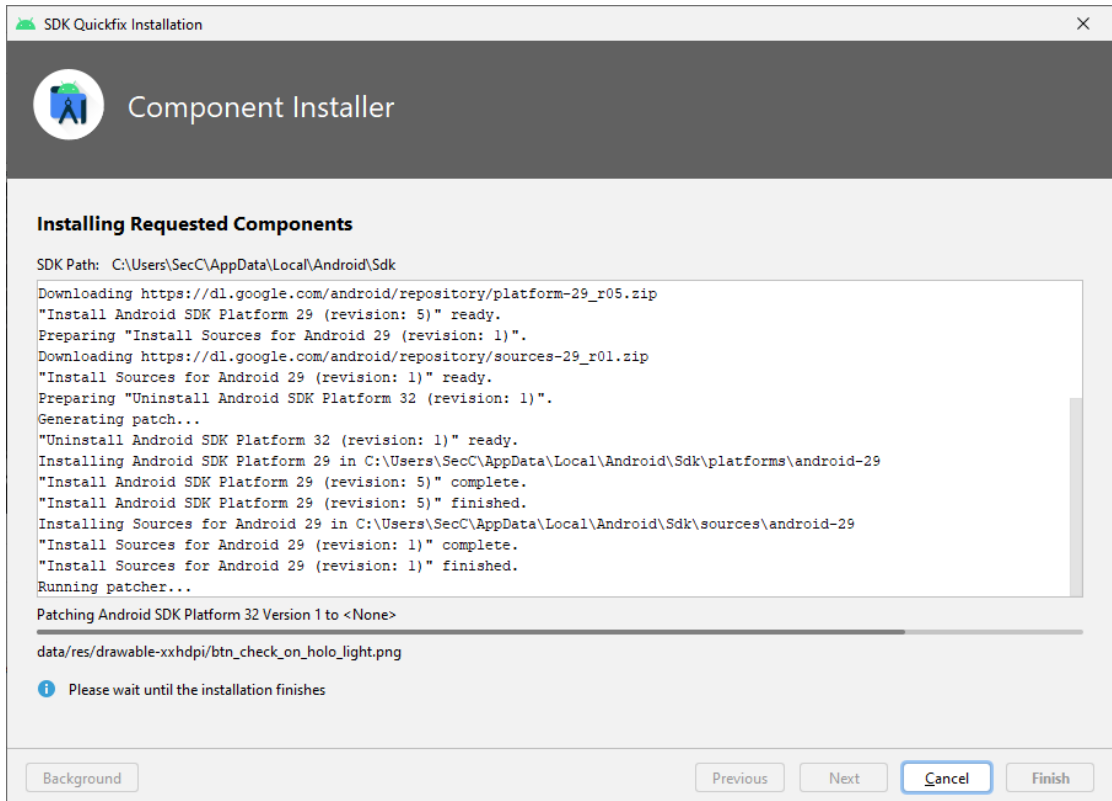
2. Start Android Studio, and go through the ‘Android Studio Setup Wizard’. This installs the latest Android SDK, Android SDK Command-line Tools, and Android SDK Build-Tools, which are required by Flutter when developing for Android.



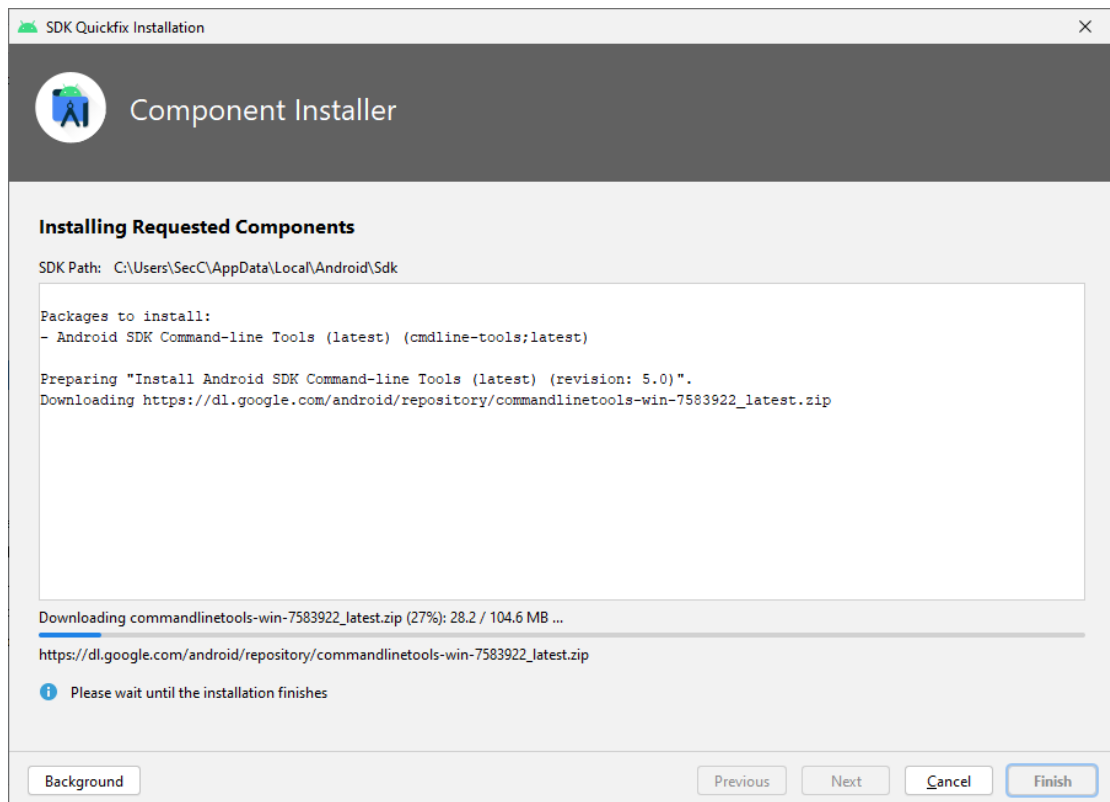
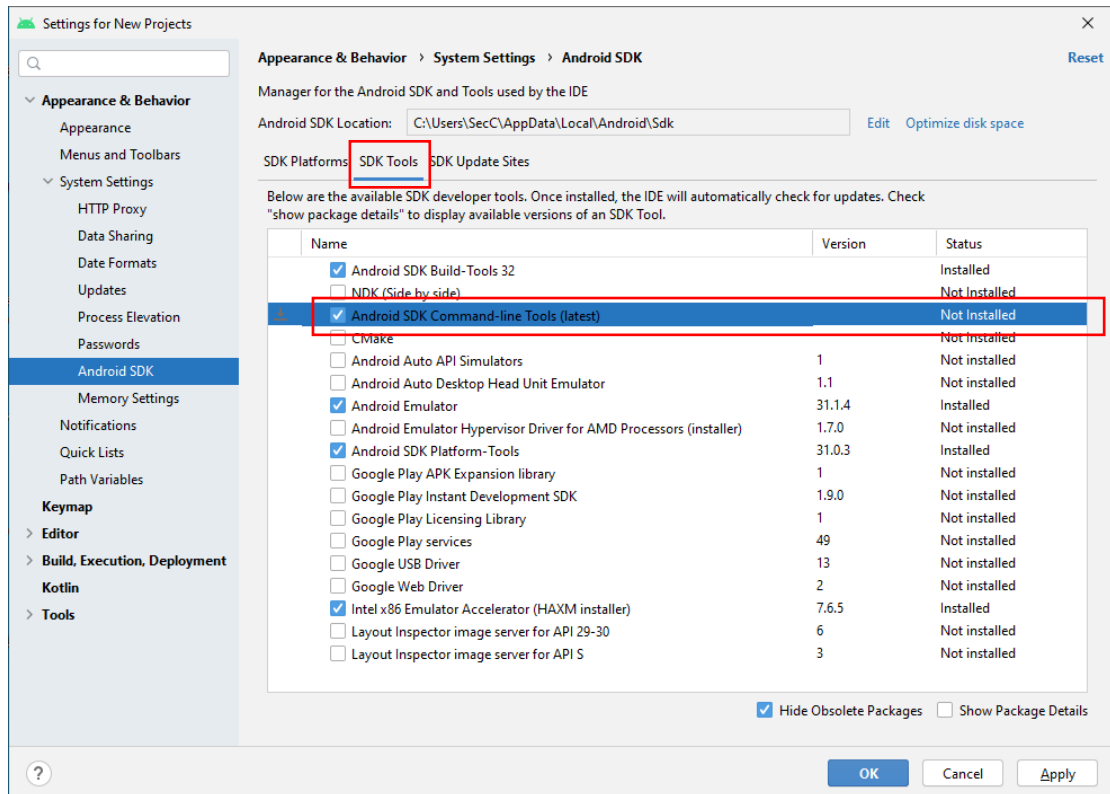
ไปที่ SDK Manager



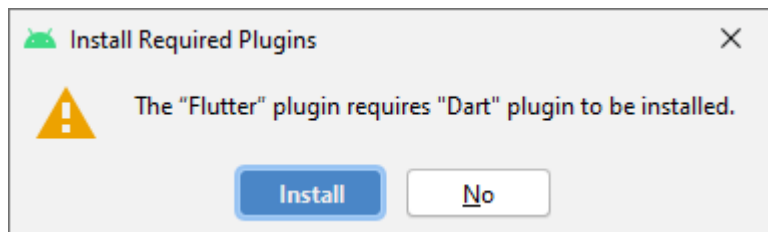
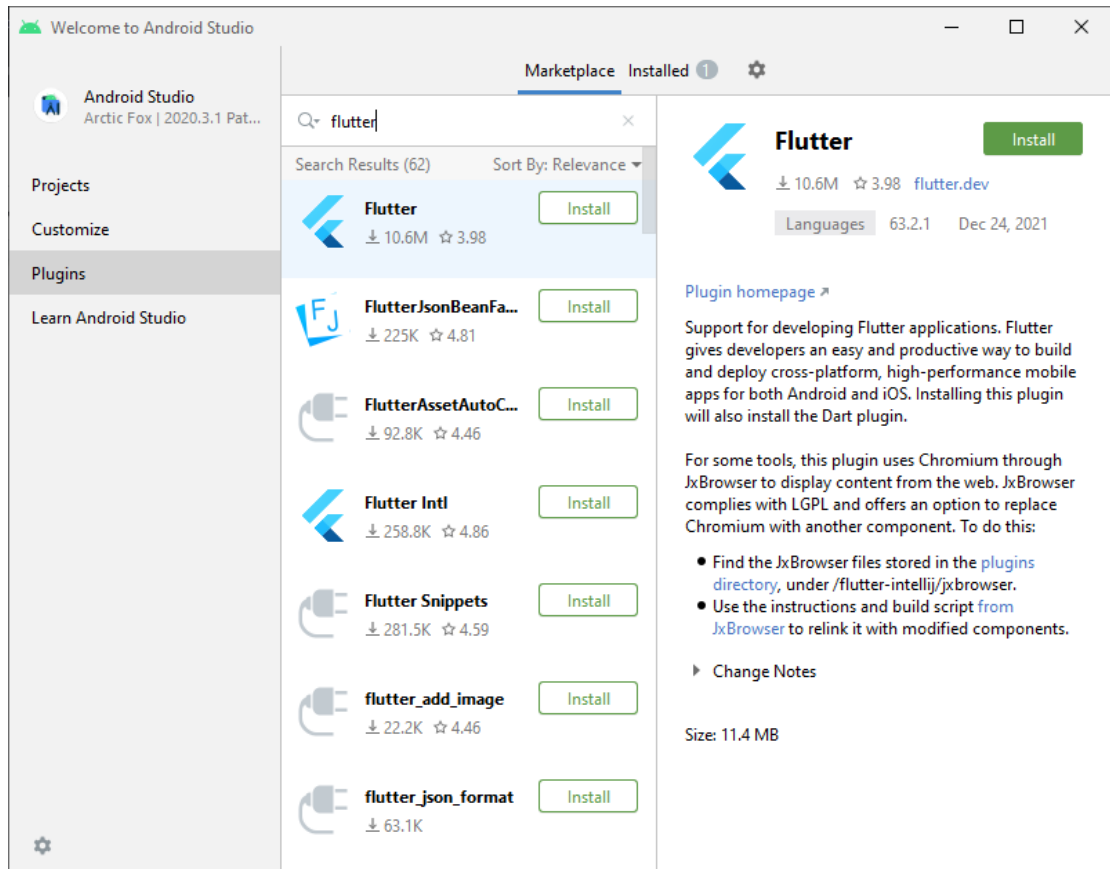




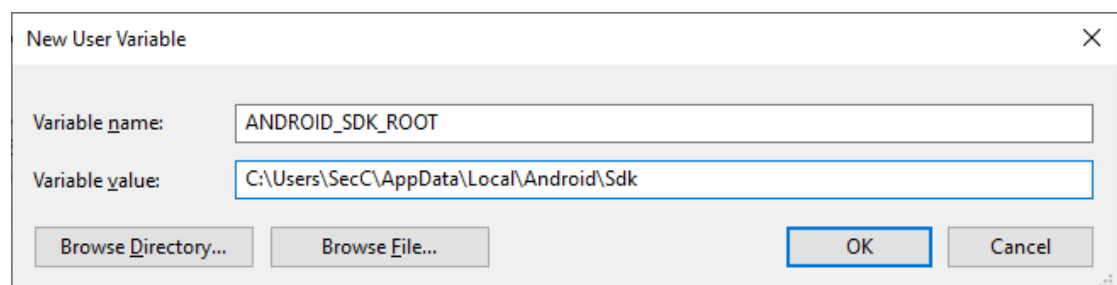
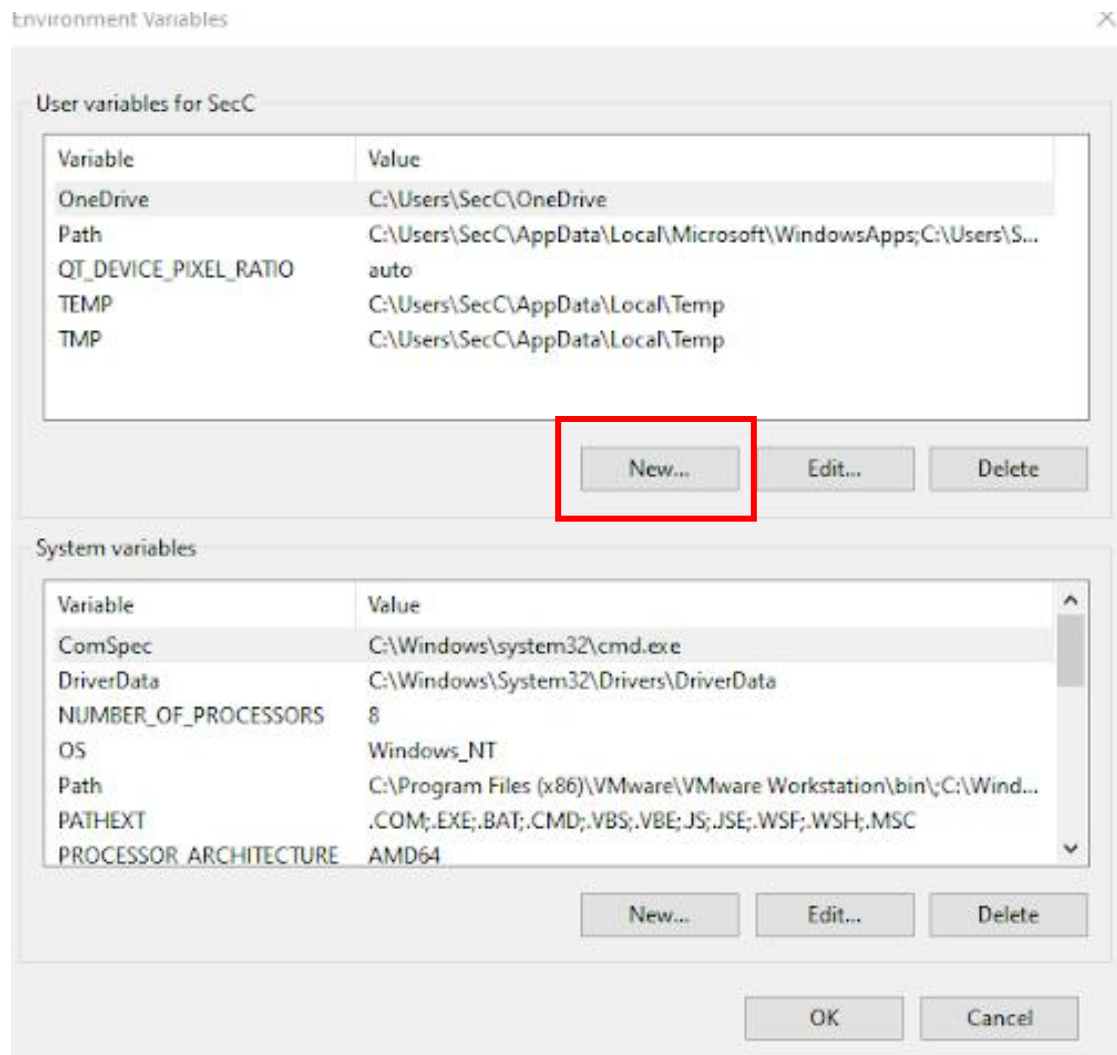
ติดตั้ง Android SDK Command-line Tools

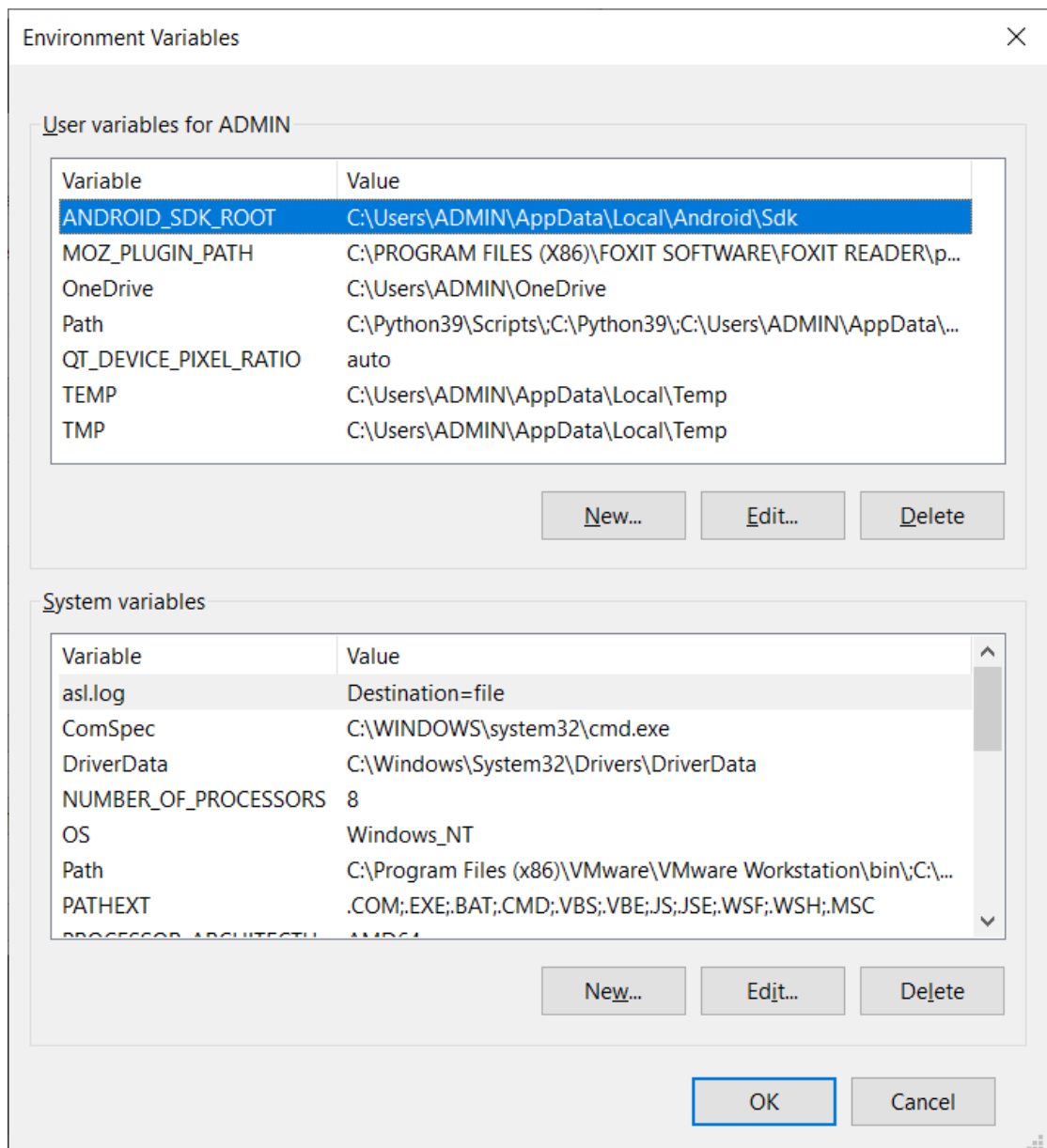


ติดตั้ง plugin Flutter และ Dart ให้กับ android studio

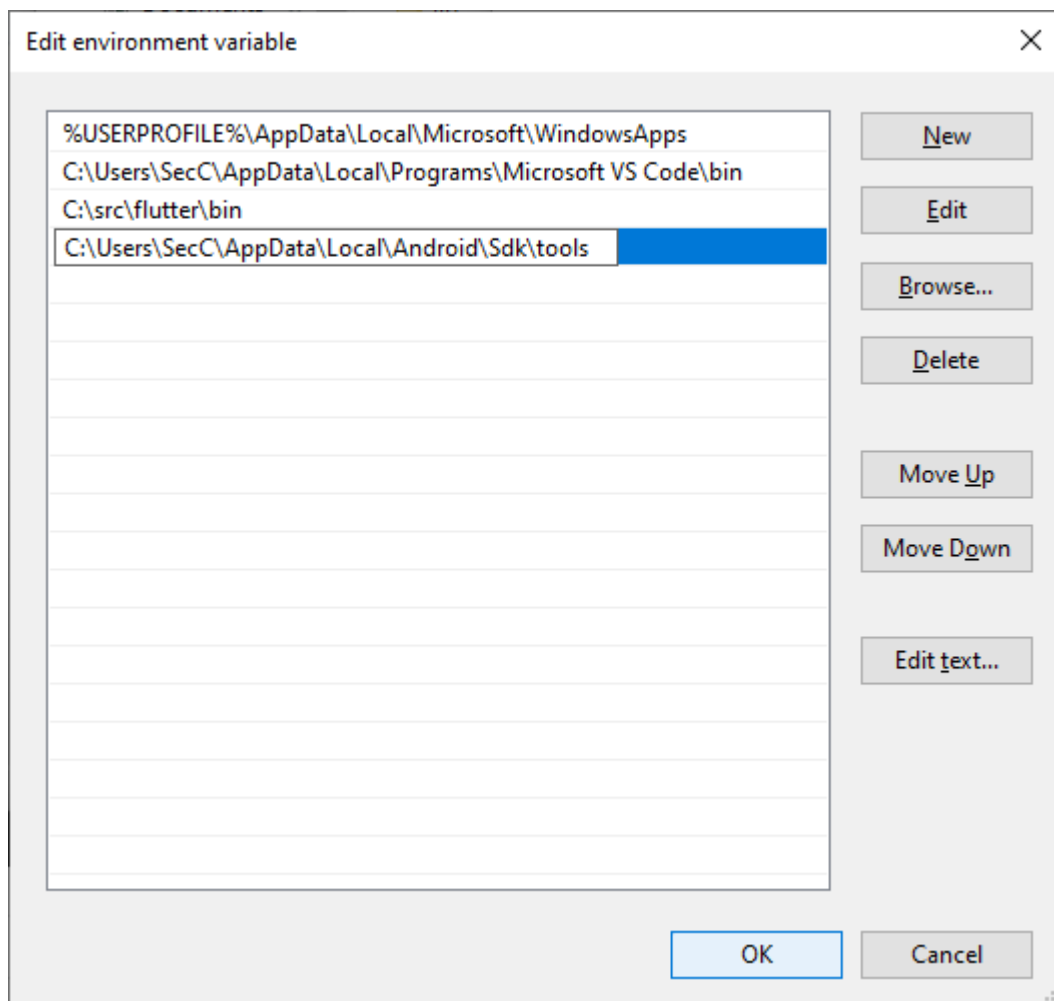
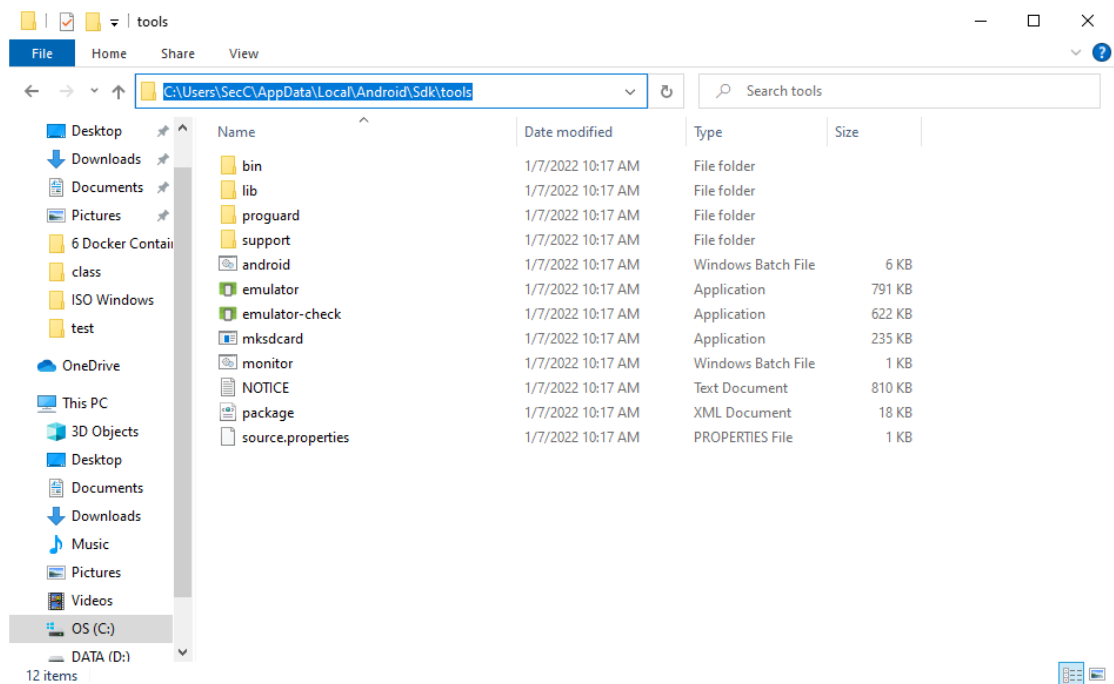


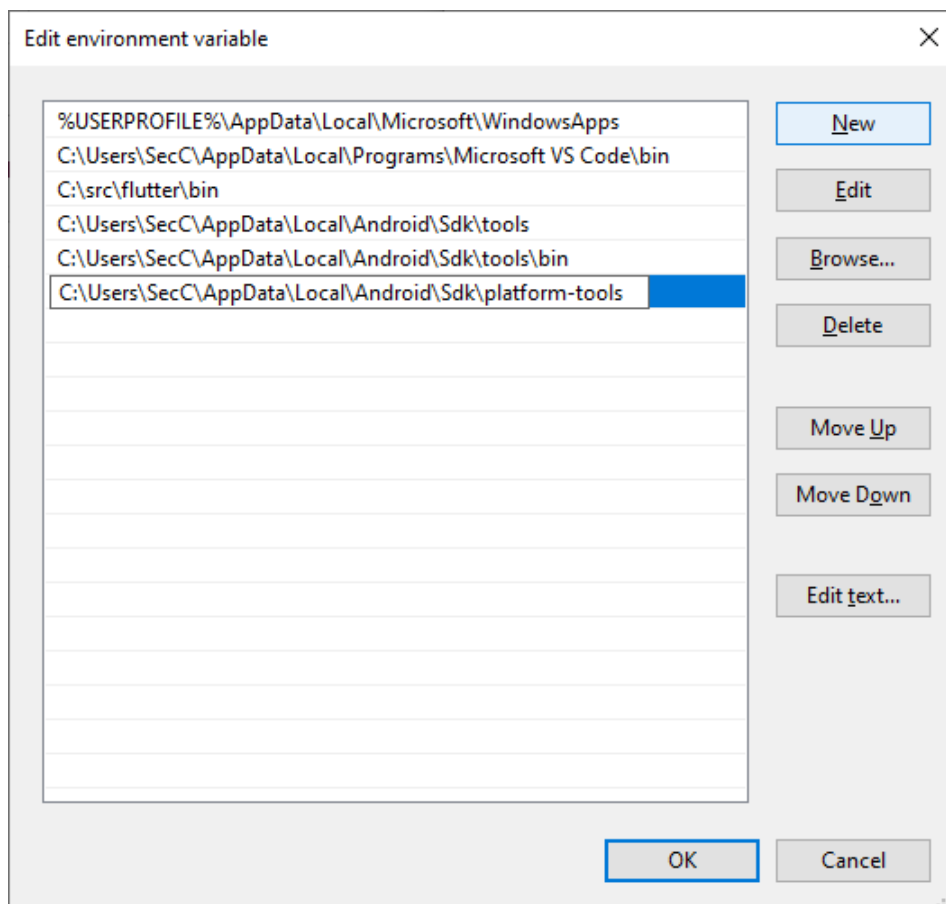
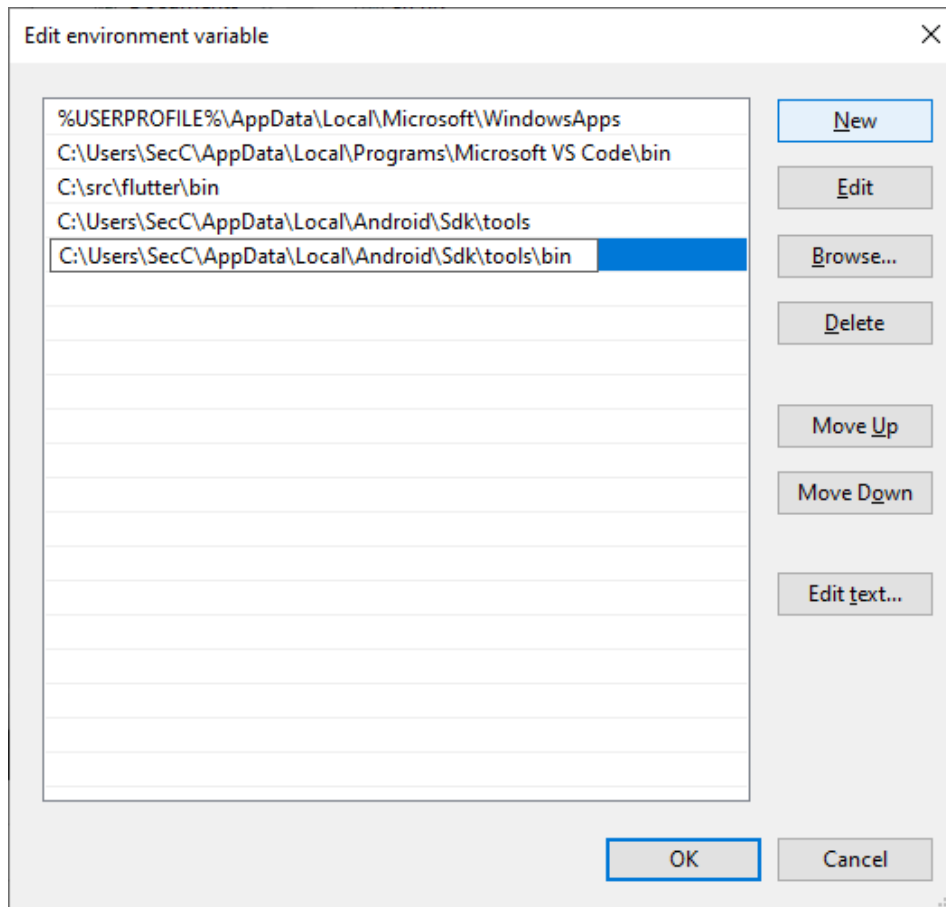
Set up your Android device





Set Path for Android SDK





C:\src\flutter>flutter doctor --android-licenses

```
Command Prompt - flutter doctor --android-licenses
C:\Users\SecC>flutter doctor --android-licenses
6 of 7 SDK package licenses not accepted. 100% Computing updates...
Review licenses that have not been accepted (y/N)?
```

ให้เลือก y เพื่อ accept

```
Command Prompt - flutter doctor --android-licenses
with all applicable laws, regulations, orders, or other restrictions of the U.S. Export Administration Regulations and the appropriate foreign government. Recipient agrees that neither it nor any of its subsidiaries will export/re-export any technical data, process, product, or service, directly or indirectly, to any country for which the United States government or any agency thereof or the foreign government from where it is shipping requires an export license, or other governmental approval, without first obtaining such license or approval. Recipient also agrees to implement measures to ensure that foreign national employees are authorized to receive any information controlled by U.S. export control laws. An export is "deemed" to take place when information is released to a foreign national wherever located.

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-----
Accept? (y/N):
```

3.Run **flutter doctor** to confirm that Flutter has located your installation of Android Studio. If Flutter cannot locate it, run **flutter config --android-studio-dir <directory>** to set the directory that Android Studio is installed to.

```
C:\Users\ADMIN>flutter config --android-studio-dir="C:\Program Files\Android\Android Studio"
```

เมื่อติดตั้งเสร็จ ให้ใช้ flutter doctor ทำการตรวจสอบองค์ประกอบอีกครั้ง

```
Command Prompt
C:\Users\ADMIN>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 2.8.1, on Microsoft Windows [Version
    10.0.19042.1415], locale en-US)
[✓] Android toolchain - develop for Android devices (Android SDK version 32.0.0)
[✓] Chrome - develop for the web
[✓] Android Studio (version 2020.3)
[!] Android Studio
    X android-studio-dir = C:\Users\ADMIN\AppData\Local\Android\Sdk
    X Unable to find bundled Java version.
[✓] VS Code (version 1.63.1)
[✓] Connected device (3 available)

! Doctor found issues in 1 category.

C:\Users\ADMIN>flutter config --android-studio-dir="C:\Program Files\Android\Android Studio"
Setting "android-studio-dir" value to "C:\Program Files\Android\Android Studio".

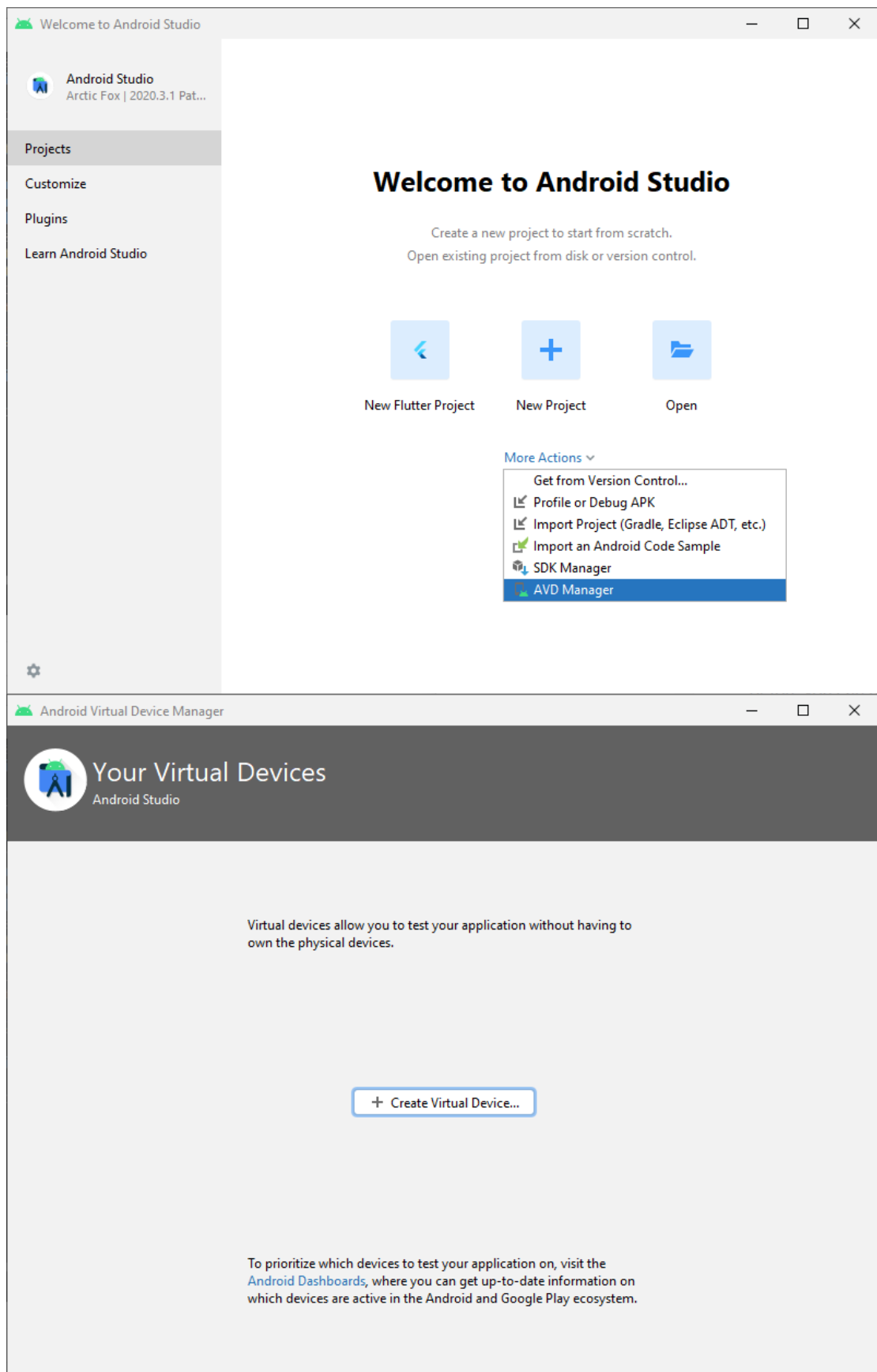
You may need to restart any open editors for them to read new settings.

C:\Users\ADMIN>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 2.8.1, on Microsoft Windows [Version
    10.0.19042.1415], locale en-US)
[✓] Android toolchain - develop for Android devices (Android SDK version 32.0.0)
[✓] Chrome - develop for the web
[✓] Android Studio (version 2020.3)
[✓] VS Code (version 1.63.1)
[✓] Connected device (3 available)

• No issues found!

C:\Users\ADMIN>flutter config --android-studio-dir="C:\Program Files\Android\Android Studio"
```

Create AVD



Virtual Device Configuration

Select Hardware

Choose a device definition

Category

TV

Phone

Wear OS

Tablet

Automotive

Name

Pixel XL

Pixel 5

Pixel 4a

Pixel 4 XL

Pixel 4

Pixel 3a XL

Pixel 3a

Pixel 3 XL

Pixel 3

Pixel 2 XL

Pixel 2

Play Store

Size

5.5"

6.0"

5.8"

6.3"

5.7"

6.0"

5.6"

6.3"

5.46"

5.99"

5.0"

Resolution

1440x2560

1080x2340

1080x2340

1440x3040

1080x2280

1080x2160

1080x2220

1440x2960

1080x2160

1440x2880

1080x1920

Density

560dpi

440dpi

440dpi

560dpi

440dpi

400dpi

440dpi

560dpi

440dpi

560dpi

420dpi

New Hardware Profile

Import Hardware Profiles

Clone Device...

Pixel 2

1080px

5.0"

1920px

Size: large

Ratio: long

Density: 420dpi

Previous

Next

Cancel

Finish

Virtual Device Configuration

System Image

Select a system image

Recommended

x86 Images

Other Images

Release Name	API Level	ABI	Target
R Download	30	x86	Android 11.0 (Google Play)
Q Download	29	x86	Android 10.0 (Google Play)
Pie Download	28	x86	Android 9.0 (Google Play)
Oreo Download	27	x86	Android 8.1 (Google Play)
Oreo Download	26	x86	Android 8.0 (Google Play)
Nougat Download	25	x86	Android 7.1.1 (Google Play)
Nougat Download	24	x86	Android 7.0 (Google Play)

Q



API Level

29

Android

10.0

Google Inc.

System Image

x86

We recommend these Google Play images because this device is compatible with Google Play.

Questions on API level?

See the [API level distribution chart](#)

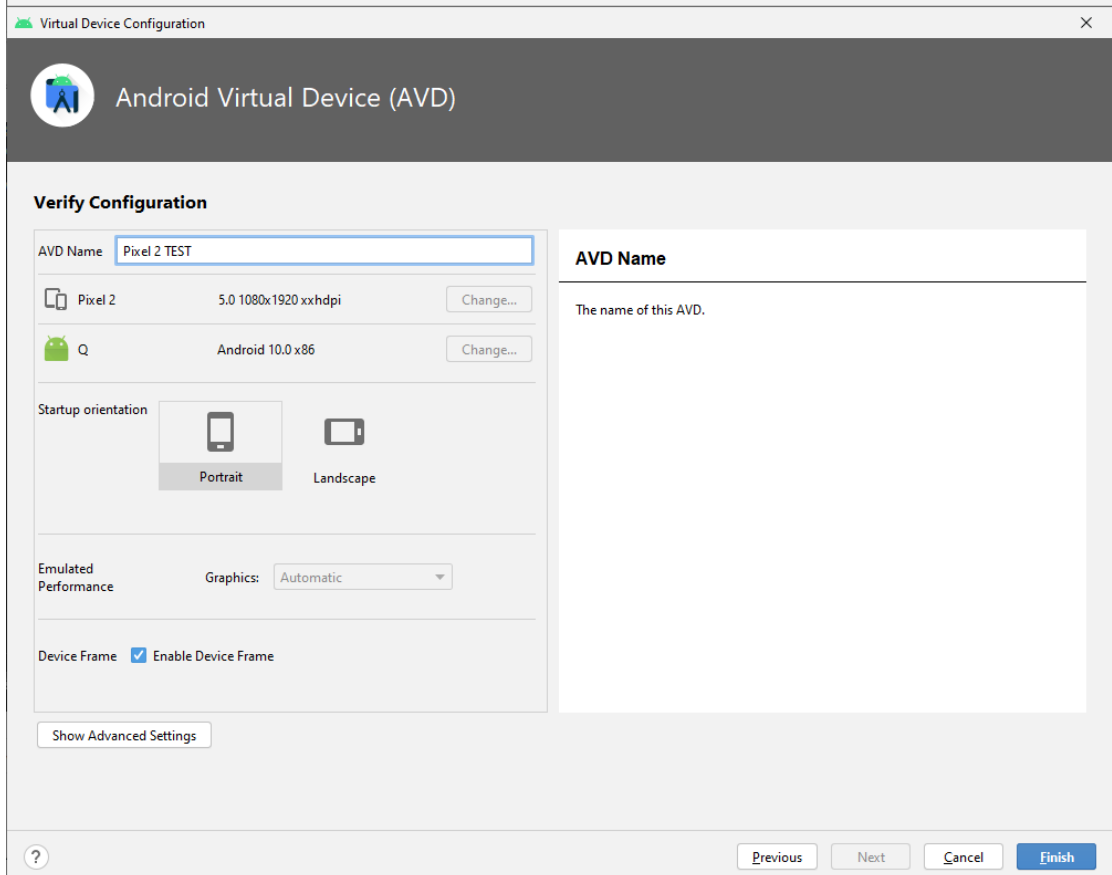
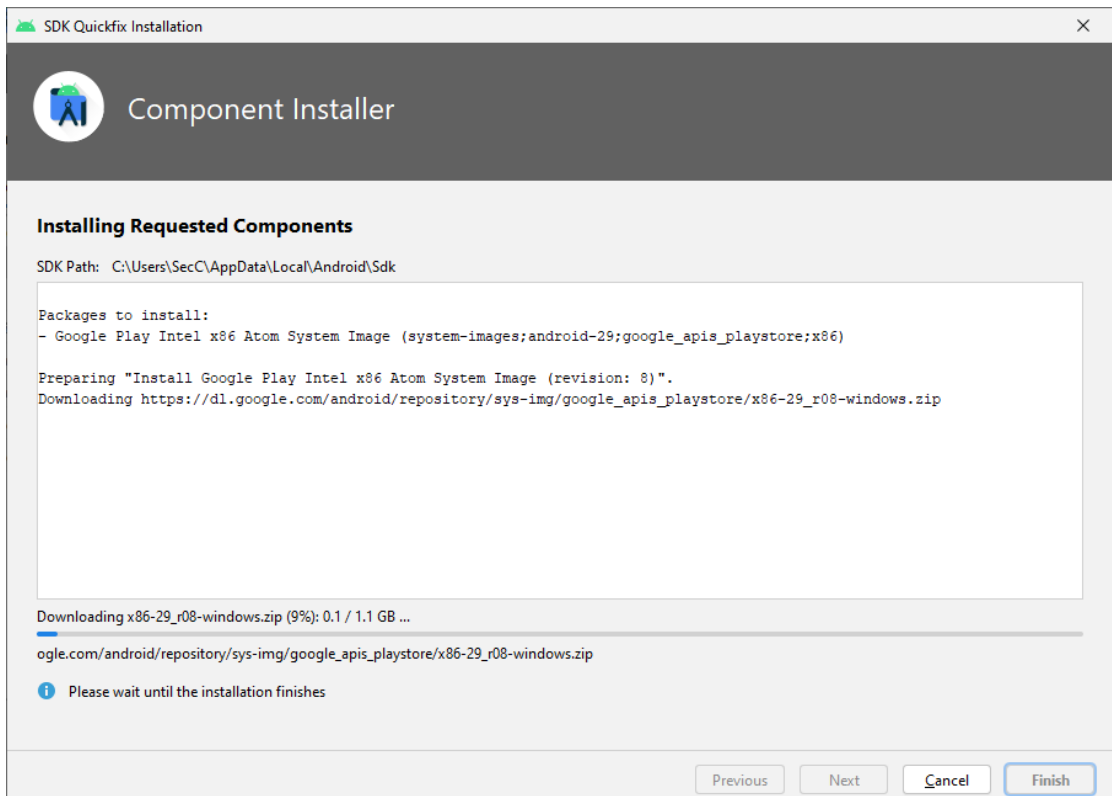
A system image must be selected to continue.

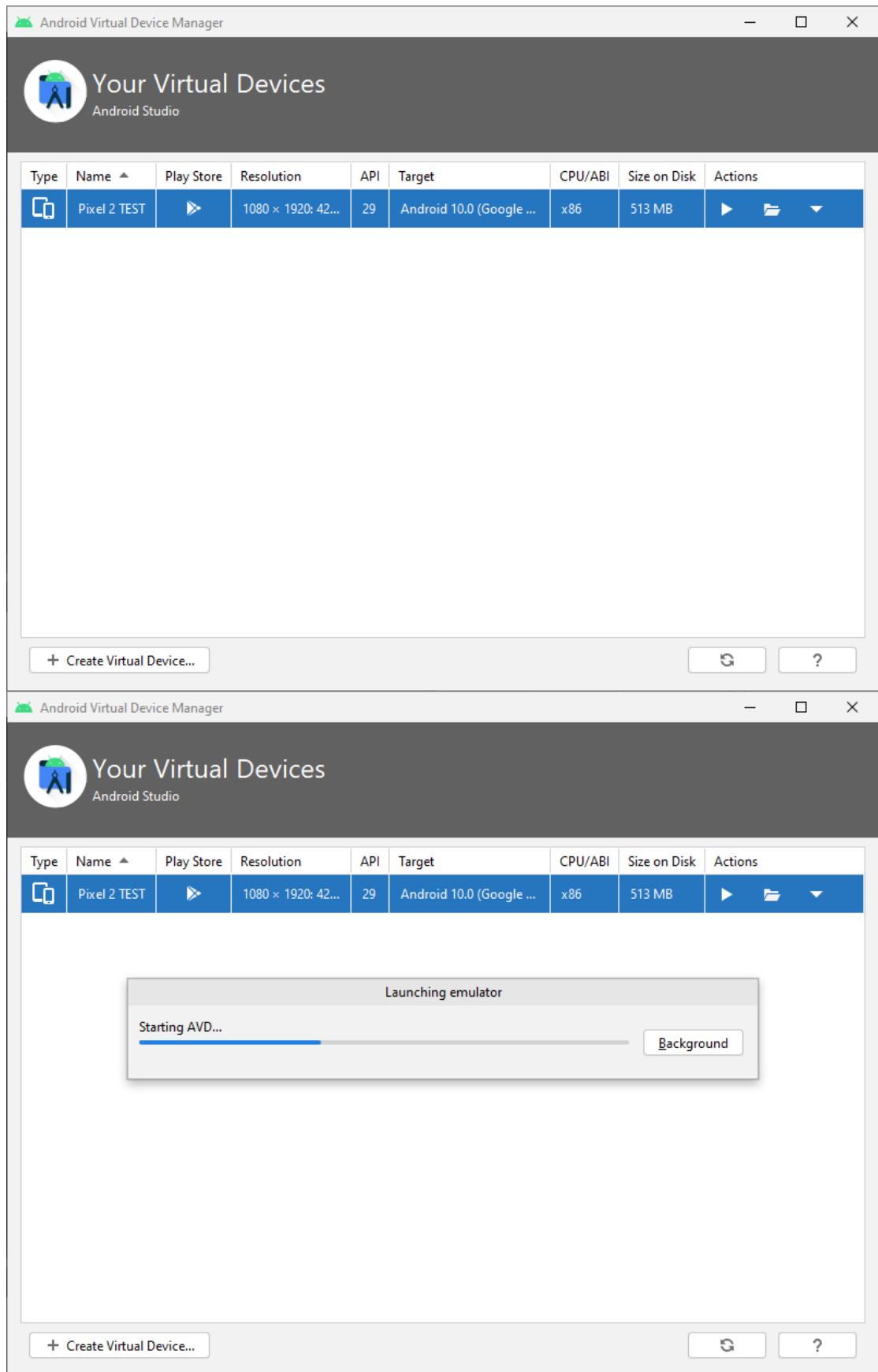
Previous

Next

Cancel

Finish







ติดตั้ง Extension ใน VS Code คือ Flutter และ Dart

The image shows two screenshots of the Visual Studio Code interface, demonstrating the installation of Flutter and Dart extensions.

Top Screenshot: Flutter Extension

- Left Panel (Extensions Marketplace):** A search for "flutter" shows several extensions. The "Flutter" extension by Dart Code is highlighted, showing it is being installed.
- Right Panel (Flutter Extension Details):** The "Flutter" extension page is displayed. It shows the Flutter logo, version v3.32.0, and a rating of 5 stars (56 reviews). The description states: "Flutter support and debugger for Visual Studio Code." The "Installing" button is visible.
- Bottom Panel (Code Editor):** A snippet of Dart code is shown, illustrating a Flutter app structure with a Scaffold, AppBar, and a Text widget.

Bottom Screenshot: Dart Extension

- Left Panel (Extensions Marketplace):** A search for "dart" shows several extensions. The "Dart" extension by Dart Code is highlighted, showing it is being installed.
- Right Panel (Dart Extension Details):** The "Dart" extension page is displayed. It shows the Dart logo, version v3.32.0, and a rating of 5 stars (54 reviews). The description states: "Dart language support and debugger for Visual Studio Code." The "Disable" and "Uninstall" buttons are visible.
- Bottom Panel (Code Editor):** A snippet of Dart code is shown, illustrating a simple Dart program with a main function.

ทดสอบการสร้าง Project ของ Flutter

วิธีที่ 1 สร้างที่ CMD

D:\MobileAPP>flutter create firstapp

```
C:\Windows\System32\cmd.exe
D:\MobileAPP>flutter create firstapp
Creating project firstapp...
  firstapp\lib\main.dart (created)
  firstapp\pubspec.yaml (created)
  firstapp\README.md (created)
  firstapp\test\widget_test.dart (created)
  firstapp\gitignore (created)
  firstapp\idea\libraries\Dart_SDK.xml (created)
  firstapp\idea\libraries\KotlinJavaRuntime.xml (created)
  firstapp\idea\modules.xml (created)
  firstapp\idea\runConfigurations\main_dart.xml (created)
  firstapp\idea\workspace.xml (created)
  firstapp\metadata (created)
  firstapp\analysis_options.yaml (created)
  firstapp\android\app\build.gradle (created)
  firstapp\android\app\src\main\kotlin\com\example\firstapp\MainActivity.kt (created)
  firstapp\android\build.gradle (created)
  firstapp\android\firstapp_android.iml (created)

  firstapp\web\icons\icon-maskable-512.png (created)
  firstapp\web\index.html (created)
  firstapp\web\manifest.json (created)
Running "flutter pub get" in firstapp... 2,464ms
Wrote 81 files.

All done!
In order to run your application, type:

  $ cd firstapp
  $ flutter run

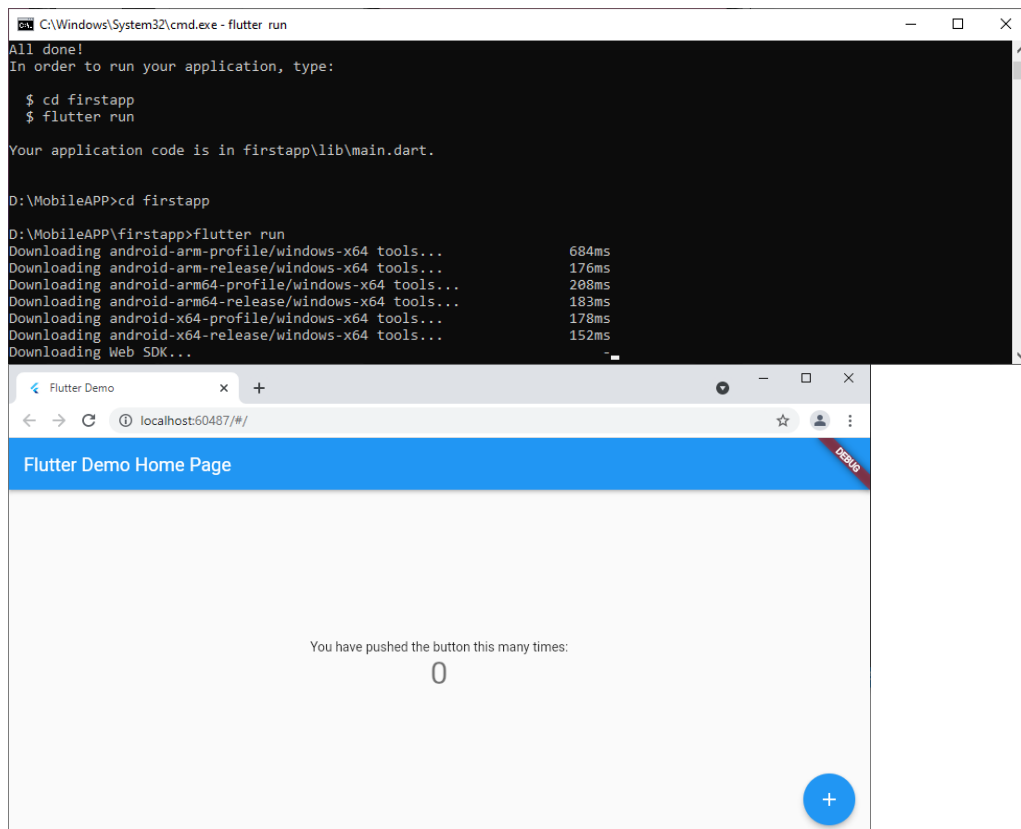
Your application code is in firstapp\lib\main.dart.

D:\MobileAPP>
```

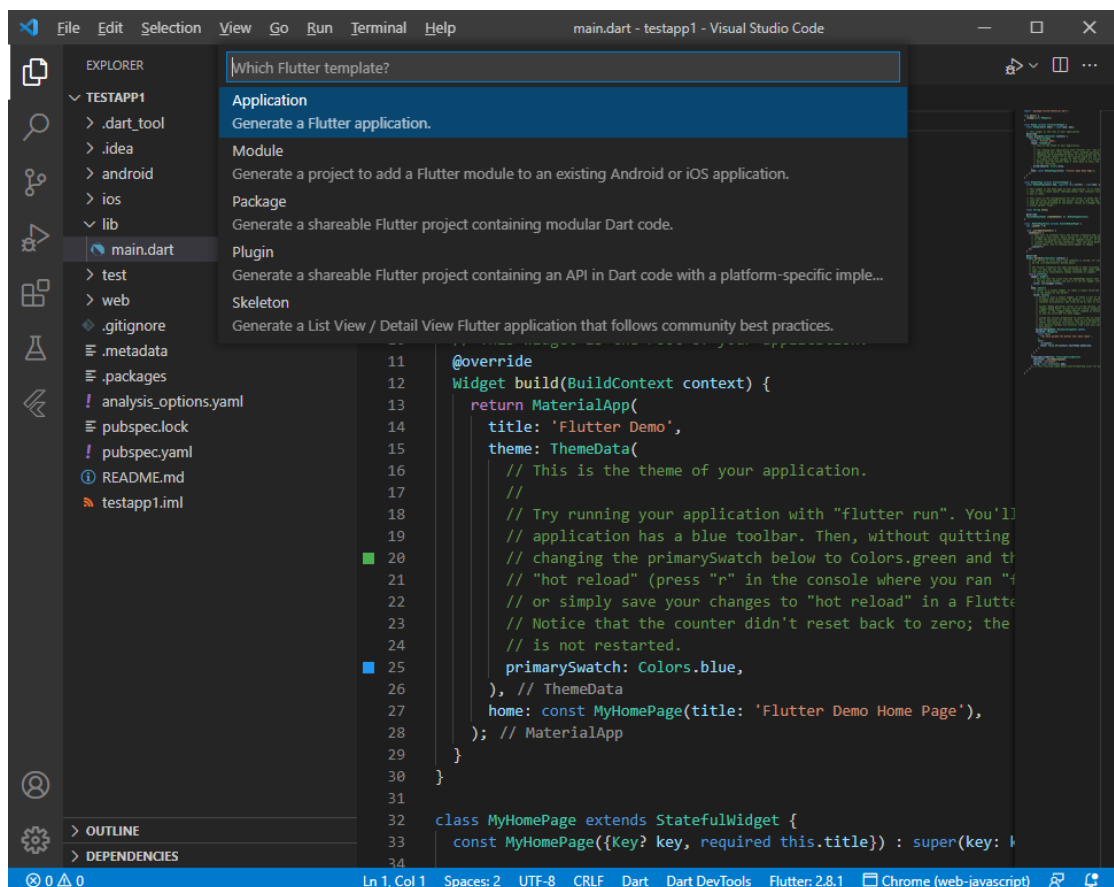
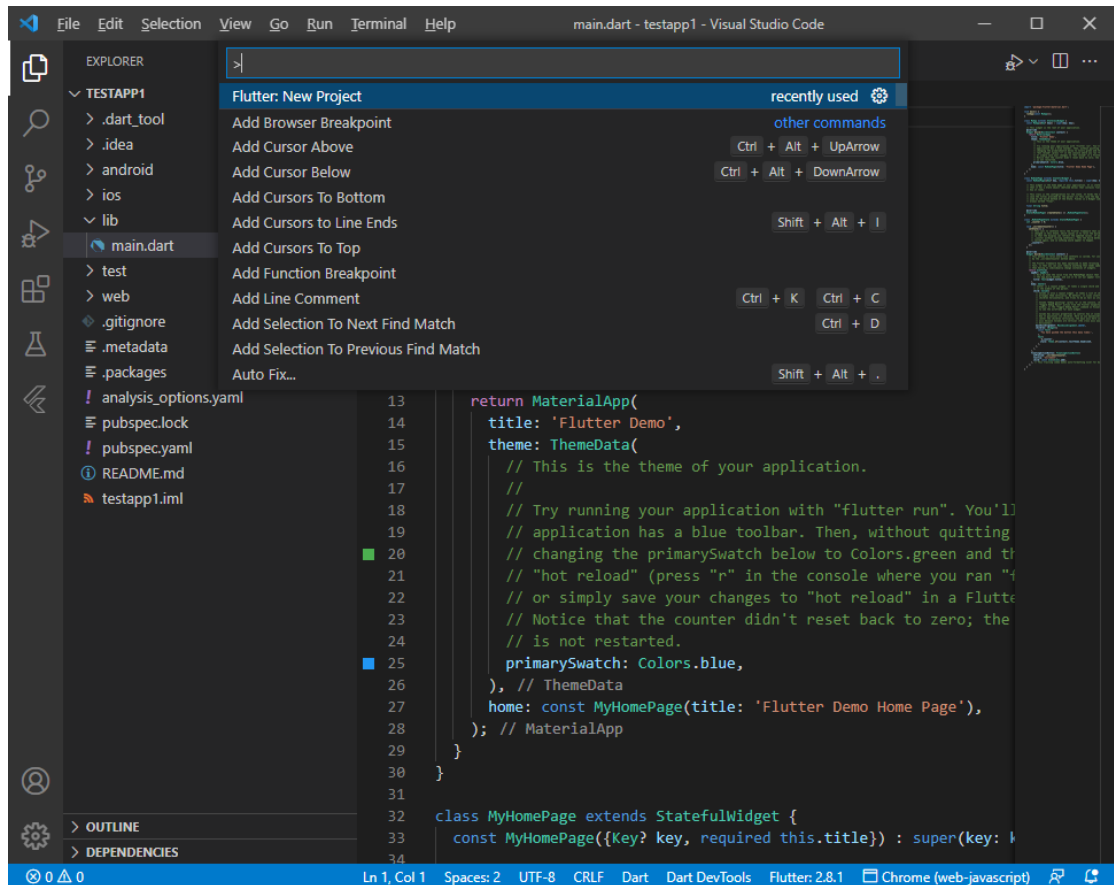
ทดสอบ การ run project

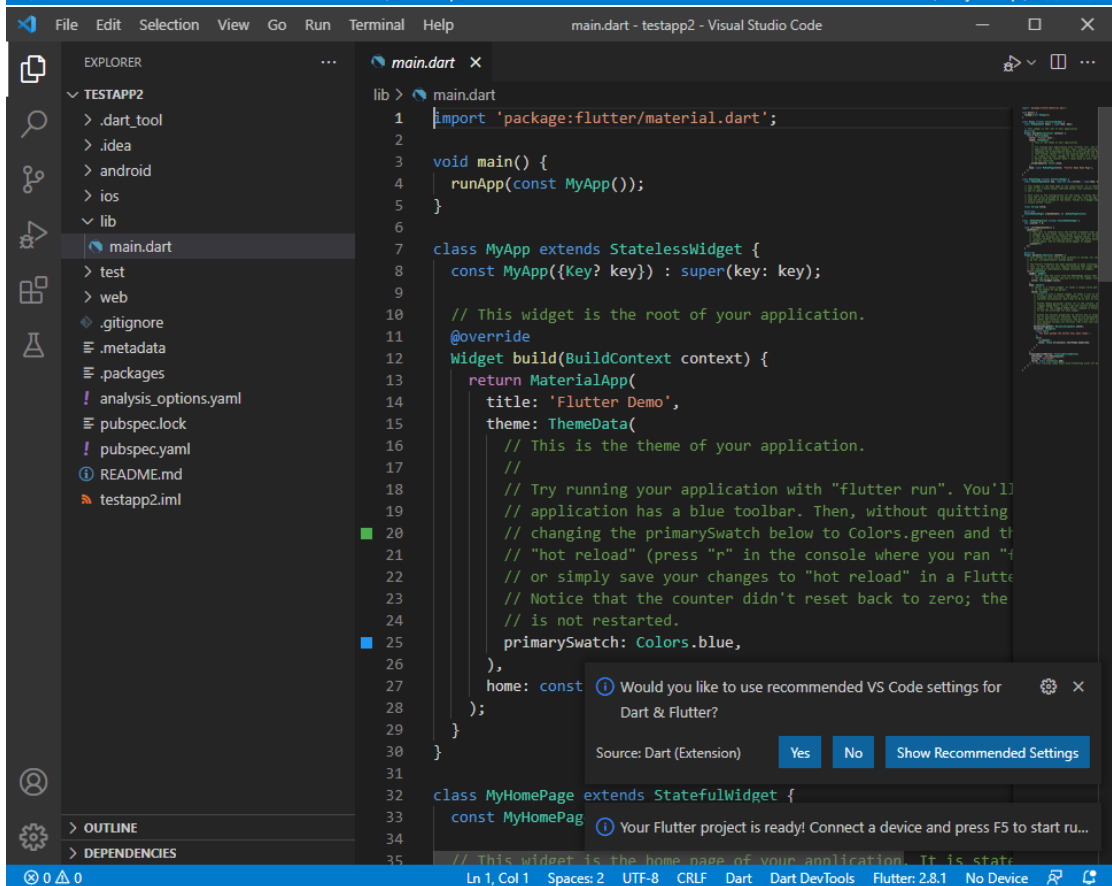
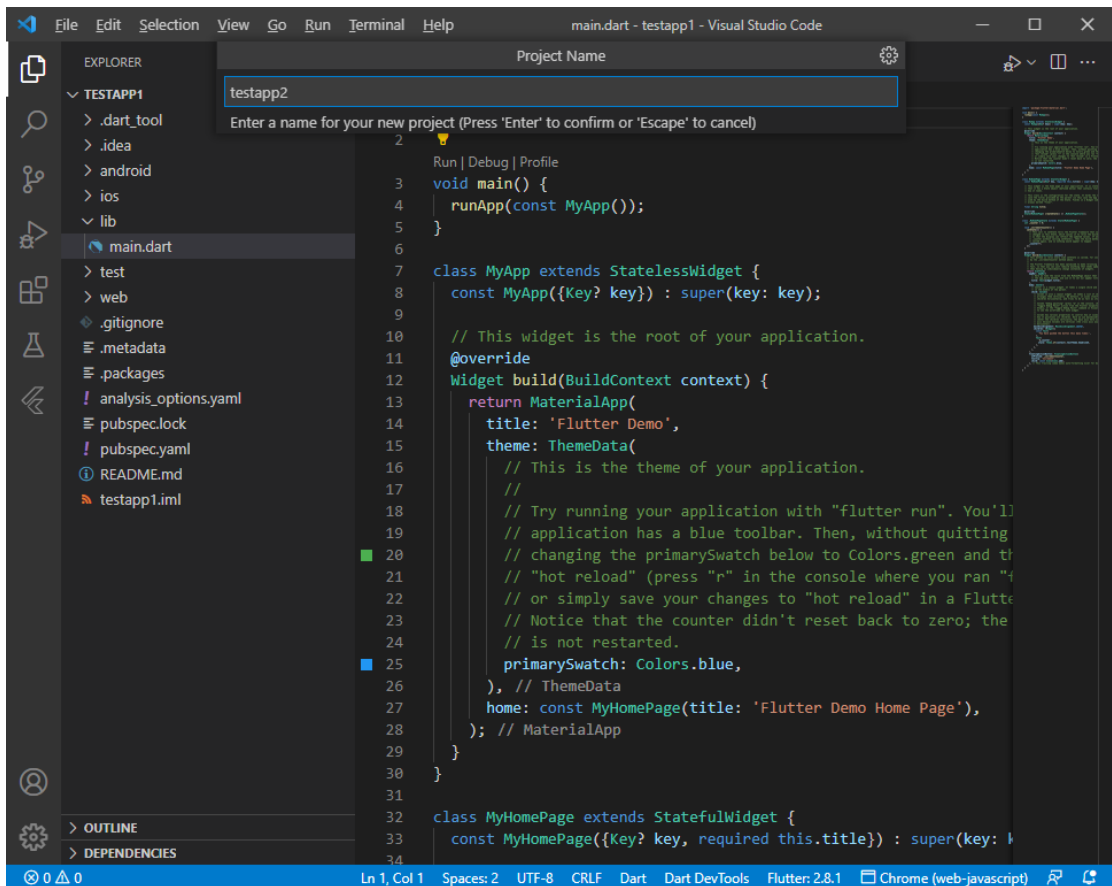
\$ cd firstapp

\$ flutter run

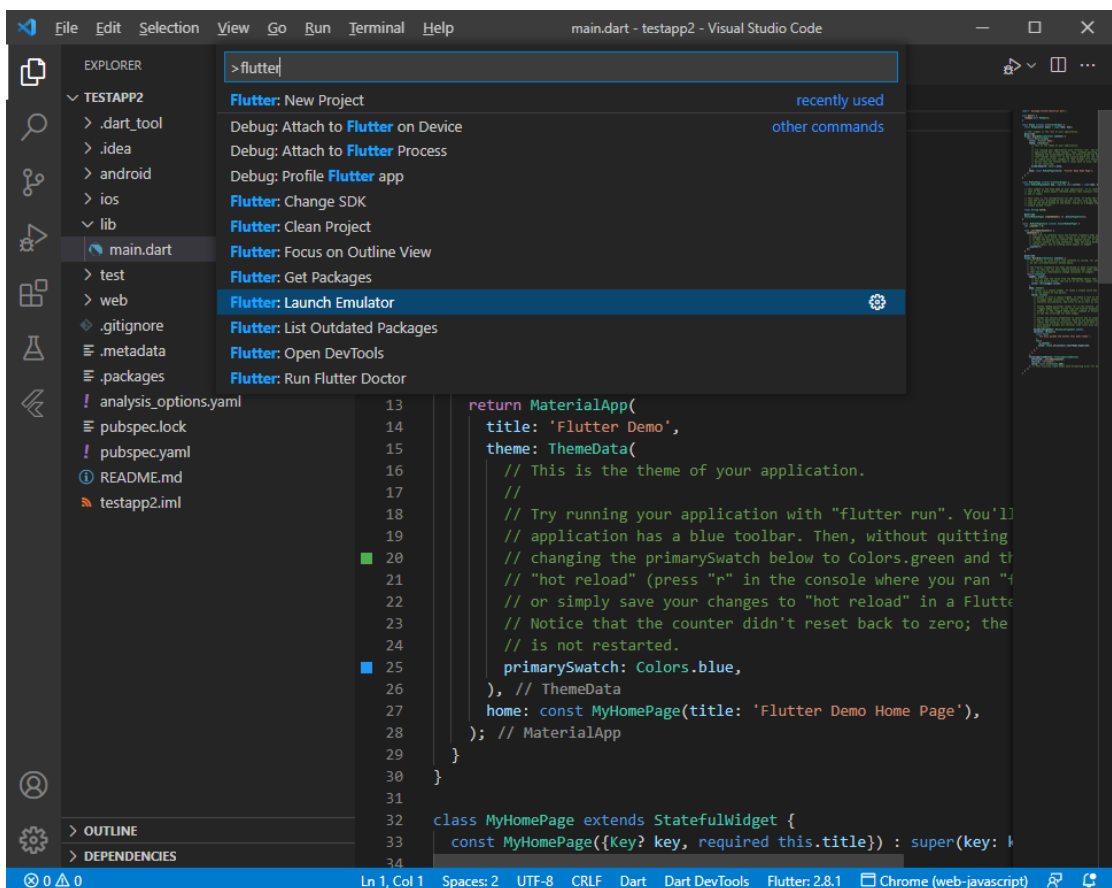
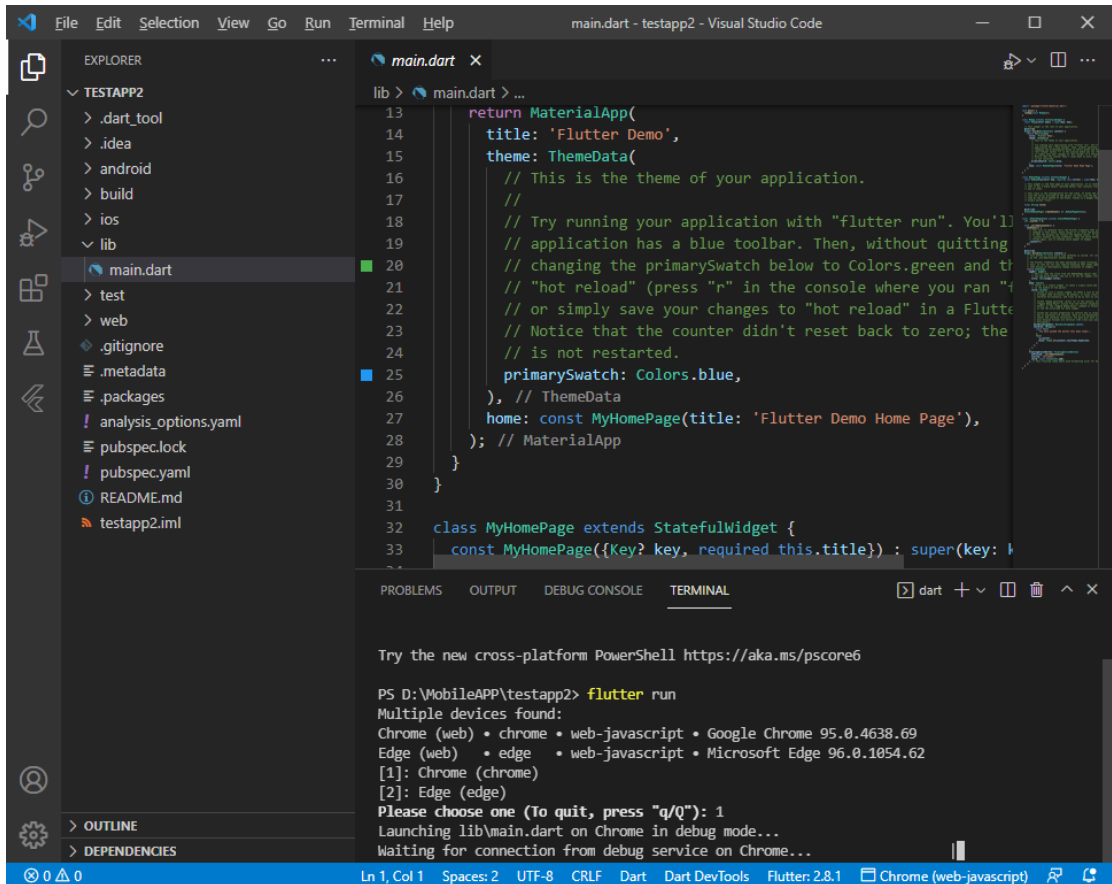


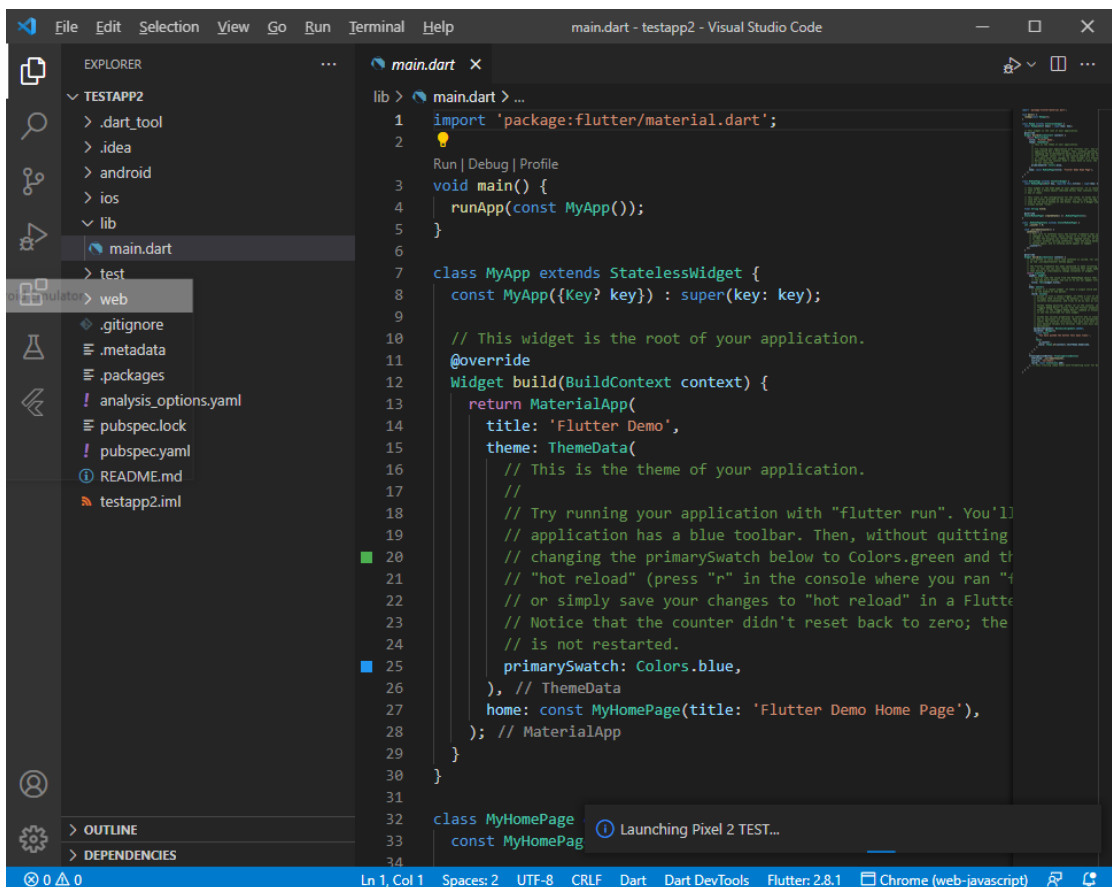
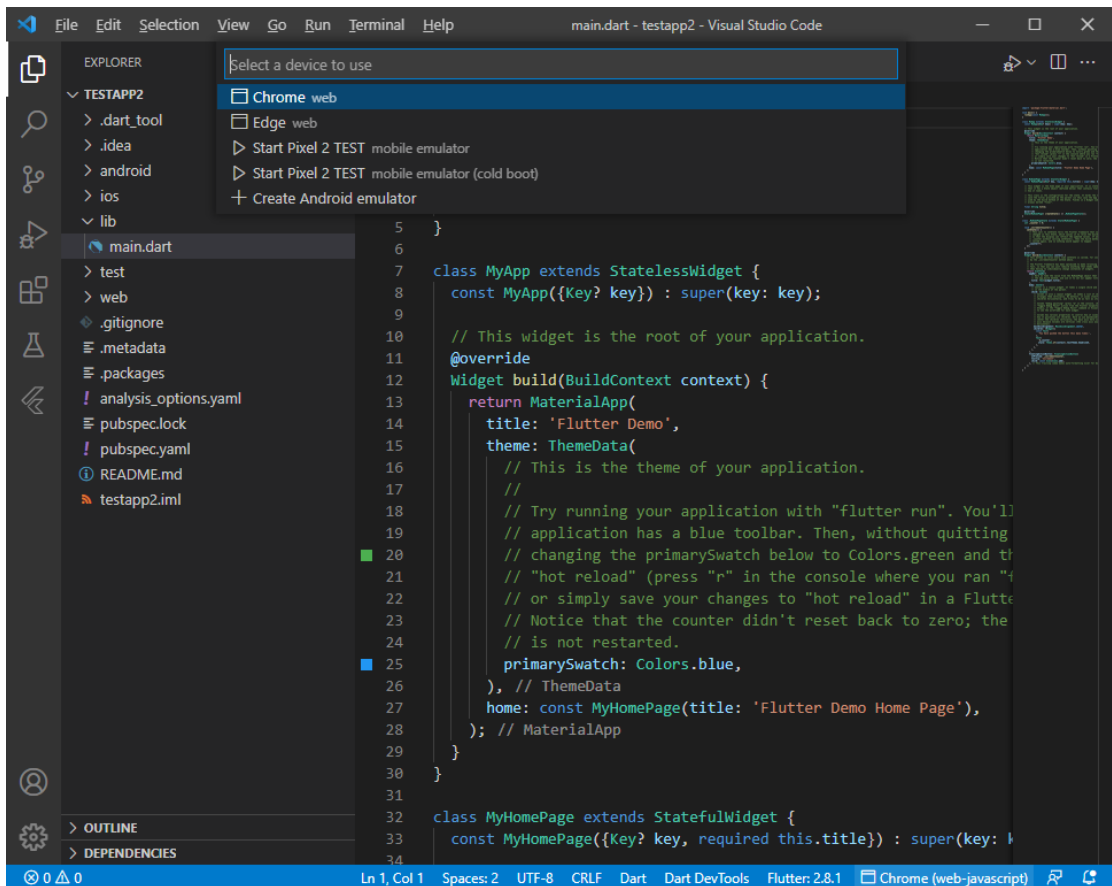
วิธีที่ 2 สร้างที่ VS Code



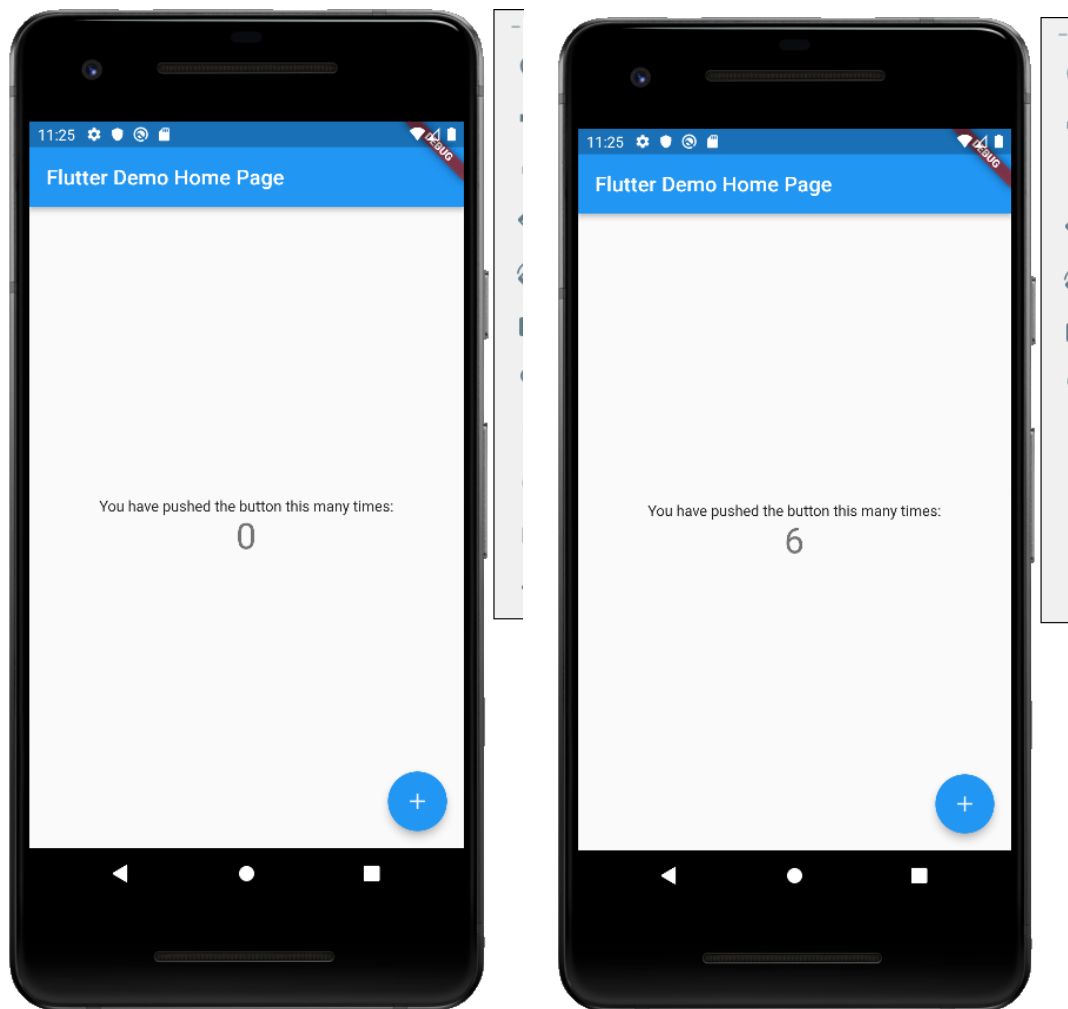


เลือก Emulator เพื่อทำการ run project





Run ด้วย AVD



Run ด้วย Web Browser

